MGSC 662: Decision Analytics

Nonlinear Optimization

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Production Planning and Pricing

Remember GTC's production planning problem:

- GTC produces wrenches and pliers!
- GTC wants to determine the daily production of wrenches and pliers so as to maximize their contribution to earnings, subject to the daily demand limits.





	Wrenches	Pliers	Availability
Required steel (lbs.)	1.5	1.0	27,000 lbs./day
Required time on molding machine (h)	1.0	1.0	21,000 h/day
Required time on assembly machine (h)	0.3	0.5	9,000 h/day
Demand limit (units/day)	15,000	16,000	
Contribution to earnings ($/unit)	$0.13	$0.10	

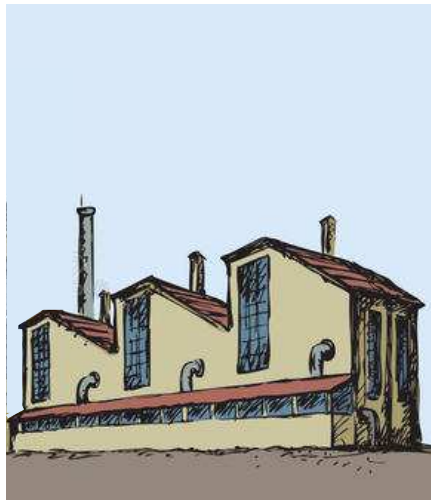
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Production Planning and Pricing

maximize	$0.13W + 0.10P$	"maximize profit"
subject to	$1.5W + 1.0P \leq 27,000$	"Availability of steel"
	$1.0W + 1.0P \leq 21,000$	"Availability of molding machine"
	$0.3W + 0.5P \leq 9,000$	"Availability of assembly machine"
	$W \leq 15,000, P \leq 16,000$	"Daily demand limits"
	$W, P \geq 0$	"We cannot produce negative quantities"

This example assumed that GTC is a price-taker:

- GTC's offering does not affect the market prices for wrenches or pliers
- Instead, GTC observes the existing market prices and
  - sells entire production if offered at or below market price
  - sells nothing if the products are offered above market price



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Production Planning and Pricing

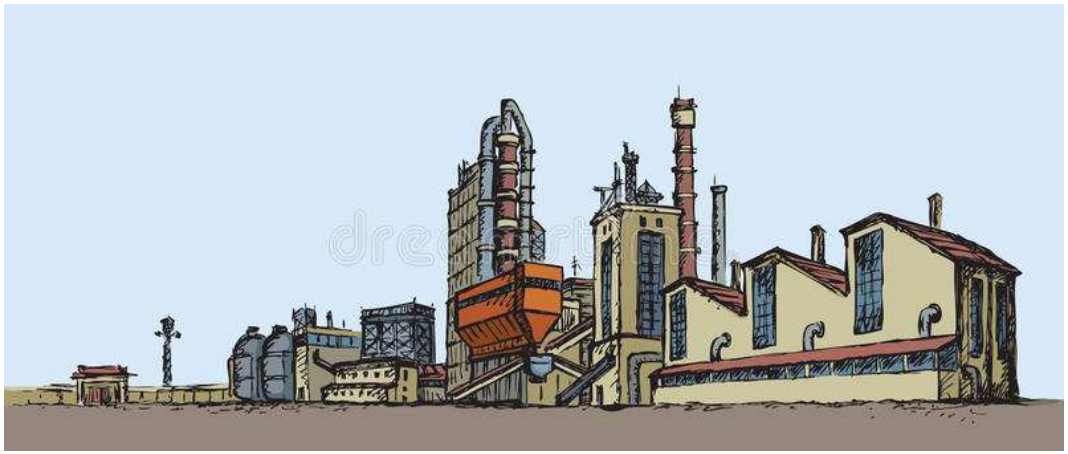
Imagine now that GTC is a price-setter:

- The market prices for wrenches or pliers depend on GTC's production

$$D_W = 565,000 - 500,000 Pr_W \qquad \text{and} \qquad D_p = 325,000 - 250,000 Pr_P$$

- In addition to the production constraints (steel, molding machine, assembly machine), GTC faces variable production costs of:

$1 per produced wrench and $1.20 per produced plier



How does the optimization model change?

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Production Planning and Pricing

GTC's new optimization problem is:

$$
\begin{aligned}
\text{maximize} \quad & W \times (Pr_W - 1.0) + P \times (Pr_P - 1.20) && \text{“profits = quantity× price”} \\
\text{subject to} \quad & 1.5W + 1.0P \leq 27,000 \\
& 1.0W + 1.0P \leq 21,000 \\
& 0.3W + 0.5P \leq 9,000 \\
& \left.\begin{aligned} W &\leq 565,000 - 500,000 Pr_W \\ P &\leq 325,000 - 250,000 Pr_P \end{aligned}\right\} && \text{quantities must not exceed demand} \\
& W, P, Pr_W, Pr_P \geq 0
\end{aligned}
$$

This is a nonlinear optimization problem!

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Production Planning and Pricing

Alternatively, set prices so that entire production is sold:

$$W \stackrel{!}{=} 565,000 - 500,000 Pr_W \implies Pr_W \stackrel{!}{=} \frac{565,000 - W}{500,000}$$

Similarly for Pr_p.

GTC's equivalent optimization problem is:

$$\begin{aligned} \text{maximize} \quad & W \times \left(\tfrac{565{,}000 - W}{500{,}000} - 1 \right) + P \times \left(\tfrac{325{,}000 - P}{250{,}000} - 1.2 \right) \\ \text{subject to} \quad & 1.5W + 1.0P \leq 27,000 \\ & 1.0W + 1.0P \leq 21,000 \\ & 0.3W + 0.5P \leq 9,000 \\ & W, P \geq 0 \end{aligned}$$

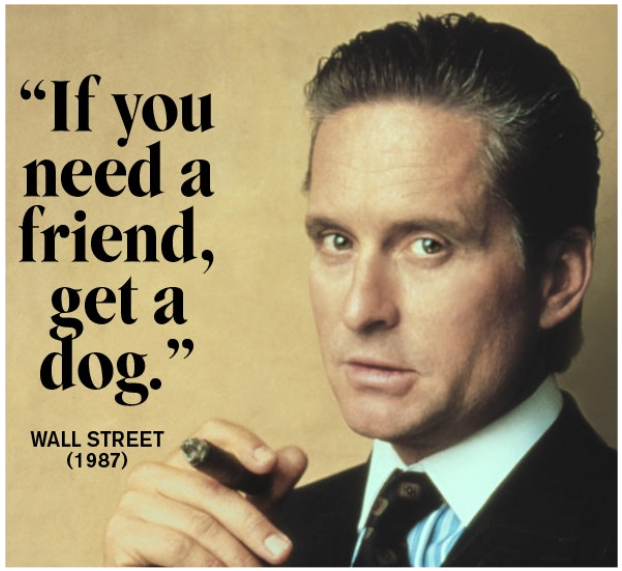
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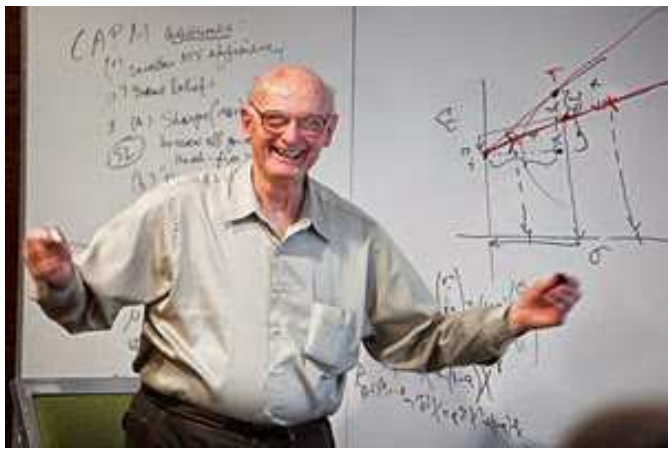
Portfolio Optimization

Investment banks and hedge funds need to select assets so as to:

- **maximize** the expected **returns**
- while **minimize** the **risks**

The goals are typically in conflict!





Harry Markowitz (Nobel Laureate 1990)

In practice, the vast majority of these companies select optimal asset portfolios using (variations of) the "**Markowitz Model**".

Consider three assets with expected return and expected standard deviation of return reported in the table. The correlation matrix is also provided.

	Annual expected return (%)	S.D. of return (%)
Asset A	11%	4 %
Asset G	14%	4.69%
Asset D	7 %	3.16%

Correlation

	Asset A	Asset G	Asset D
Asset A	1		
Asset G	0.16	1	
Asset D	-0.395	0.067	1

- Formulate an NLP to minimize risk with an expected rate of return on the portfolio of at least 11%.
- Formulate an NLP to maximize return on the portfolio with a risk of the portfolio not exceeding 3.1%.

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Define A, G, and D as the fraction of investment in each of the assets respectively. Obviously, it must be that $A + G + D = 1$.

- The return on the portfolio is $R = A \times R_A + G \times R_G + D \times R_D$. But the return of each asset is random and so R is random too. We have:

$$E(R) = 11A + 14G + 7D.$$

- What is the risk of the portfolio? Standard deviation of the portfolio is a measure of its risk. Standard deviation is the square root of the variance:

$$\begin{aligned}
\sigma_R^2 &= A^2\sigma_A^2 + G^2\sigma_G^2 + D^2\sigma_D^2 \\
&\quad +2AG\sigma_A\sigma_G Corr(R_A, R_G) + 2AD\sigma_A\sigma_D Corr(R_A, R_D) \\
&\quad +2GD\sigma_G\sigma_D Corr(R_G, R_D) \\
&= 16A^2 + 22G^2 + 10G^2 + 6AG - 10AD + 2GD.
\end{aligned}$$

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The NLP to minimize risk while guaranteeing an average return of at least 11% is:

$$
\begin{aligned}
\text{minimize} \quad & \sqrt{16A^2 + 22G^2 + 10D^2 + 6AG - 10AD + 2GD} \\
\text{subject to} \quad & \\
\text{Fractions} \quad & A + G + D = 1 \\
\text{Target return} \quad & 11A + 14G + 7D \geq 11 \\
\text{Nonnegativity} \quad & A, G, D \geq 0
\end{aligned}
$$

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The NLP to minimize risk while guaranteeing an average return of at least 11% is:

$$\begin{aligned}
\text{minimize} \quad & \sqrt{16A^2 + 22G^2 + 10D^2 + 6AG - 10AD + 2GD} \\
\text{subject to} \quad & \\
\text{Fractions} \quad & A + G + D = 1 \\
\text{Target return} \quad & 11A + 14G + 7D \geq 11 \\
\text{Nonnegativity} \quad & A, G, D \geq 0
\end{aligned}$$

The NLP to maximize return while guaranteeing a risk of at most 3.1% is:

$$\begin{aligned}
\text{maximize} \quad & 11A + 14G + 7D \\
\text{subject to} \quad & \\
\text{Fractions} \quad & A + G + D = 1 \\
\text{Target risk} \quad & \sqrt{16A^2 + 22G^2 + 10D^2 + 6AG - 10AD + 2GD} \leq 3.1 \\
\text{Nonnegativity} \quad & A, G, D \geq 0
\end{aligned}$$

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Portfolio Optimization: General Form

We want to select among $i = 1, \ldots, n$ assets with prices

$$\begin{cases} X_i & \text{today (known)} \\ \widetilde{Y}_i & \text{in the next period (random)} \end{cases}$$

We are interested in the (random) **rates of return** of those assets:

$$\widetilde{r}_i = \frac{\widetilde{Y}_i - X_i}{X_i} \in [-1, \infty)$$

Assume we have access to the following data:

- expected asset return rates: $\mu_i = E(\widetilde{r}_i)$
- covariances of asset return rates:

$$\sigma_{ij} = \mathsf{COV}(\widetilde{r}_i, \widetilde{r}_j) = E[(\widetilde{r}_i - \mu_i)(\widetilde{r}_j - \mu_j)]$$

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Portfolio Optimization: The Measures

Assume we invest a fraction $x_i \in [0, 1]$ into each asset $i = 1, \ldots, n$:

Expected Portfolio Return: measures the **expected returns**

$$E\left(\sum_{i=1}^{n} \widetilde{r}_i x_i\right) = \sum_{i=1}^{n} E(\widetilde{r}_i) x_i = \mu^\top x$$

Portfolio Variance: measures the **risk**

$$\begin{aligned} \mathsf{VAR}\left(\sum_{i=1}^{n} \widetilde{r}_i x_i\right) &= \sum_{i=1}^{n}\sum_{j=1}^{n} \mathsf{COV}(\widetilde{r}_i x_i, \widetilde{r}_j x_j) \\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} \mathsf{COV}(\widetilde{r}_i, \widetilde{r}_j) x_i x_j = x^\top \Sigma x \end{aligned}$$

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Portfolio Optimization: The Measures

Return Maximization Formulation:

maximize	$\mu^\top x$	"maximize expected returns"
subject to	$\mathrm{e}^\top x = 1$	"budget constraint"
	$x^\top \Sigma x \leq \bar{R}$	"maximum acceptable risk"
	$x \geq 0$	"no short selling"

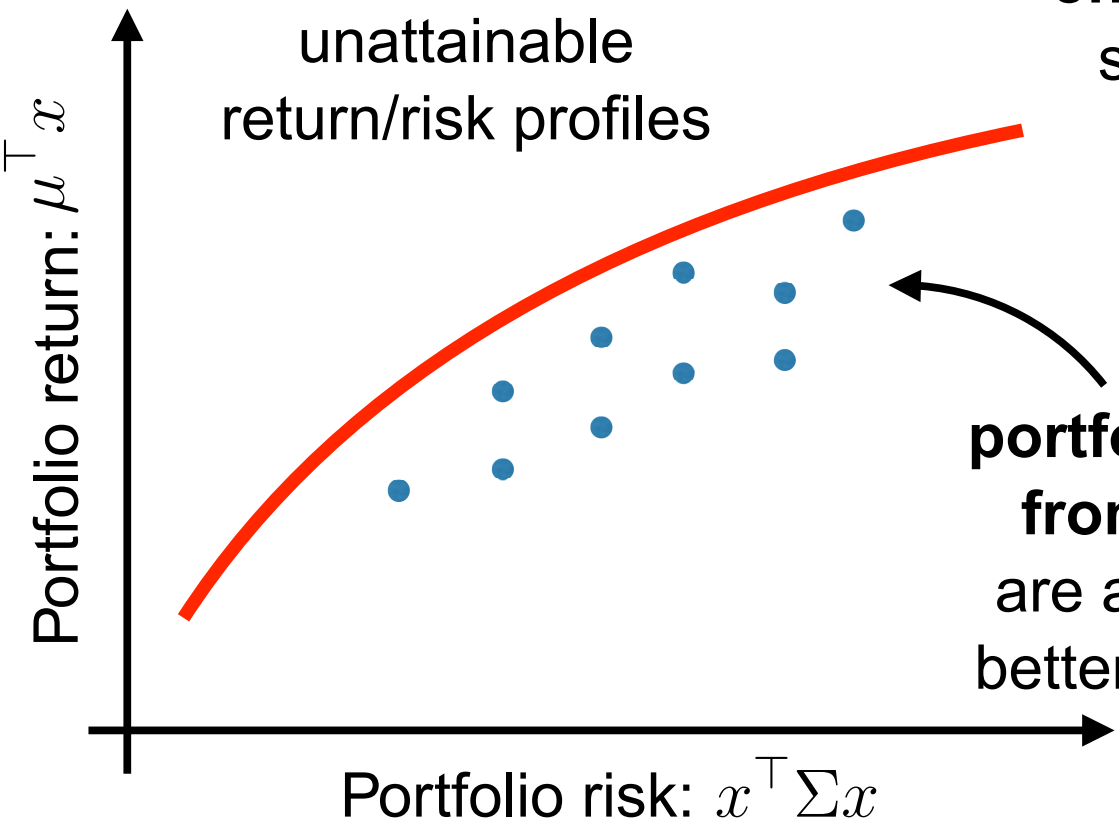
Risk Minimization Formulation:

minimize	$x^\top \Sigma x$	"minimize portfolio risk"
subject to	$\mathrm{e}^\top x = 1$	"budget constraint"
	$\mu^\top x \geq \underline{R}$	"minimum acceptable return"
	$x \geq 0$	"no short selling"

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Portfolio Optimization: The Efficient Frontier

Solving either model parametrically:



efficient frontier formed by solutions of either model

portfolios below the efficient frontier are inferior: there are alternative portfolios with better return and/or risk profile

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Portfolio Optimization: Practical Extensions

- **Minimum/maximum investment levels for individual assets**

$$\underline{x}_i \leq x_i \leq \overline{x}_i \quad \text{for some or all } i = 1, \ldots, n$$

- **Minimum/maximum investment levels for asset groups**

$$\underline{x}_J \leq \sum_{i \in J} x_i \leq \overline{x}_J \quad \text{for some } J \subseteq 1, \ldots, n$$

- **Transaction costs**

$$\text{subtract } \sum_{i=1}^{n} |x_i - x_i^0| \text{ or } \sum_{i=1}^{n} (x_i - x_i^0)^2 \quad \text{from expected return}$$

  where x_i^0, $i = 1, \ldots, n$, is the currently held portfolio.

- **Cardinality constraints** (see Discrete Optimization lecture)

$$|\{i : x_i > 0\}| \leq K$$

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Solving Non-Linear Optimization Problems:

Graphical Solution

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Solving Non-Linear Programs: Graphical Solution

Consider the problem:

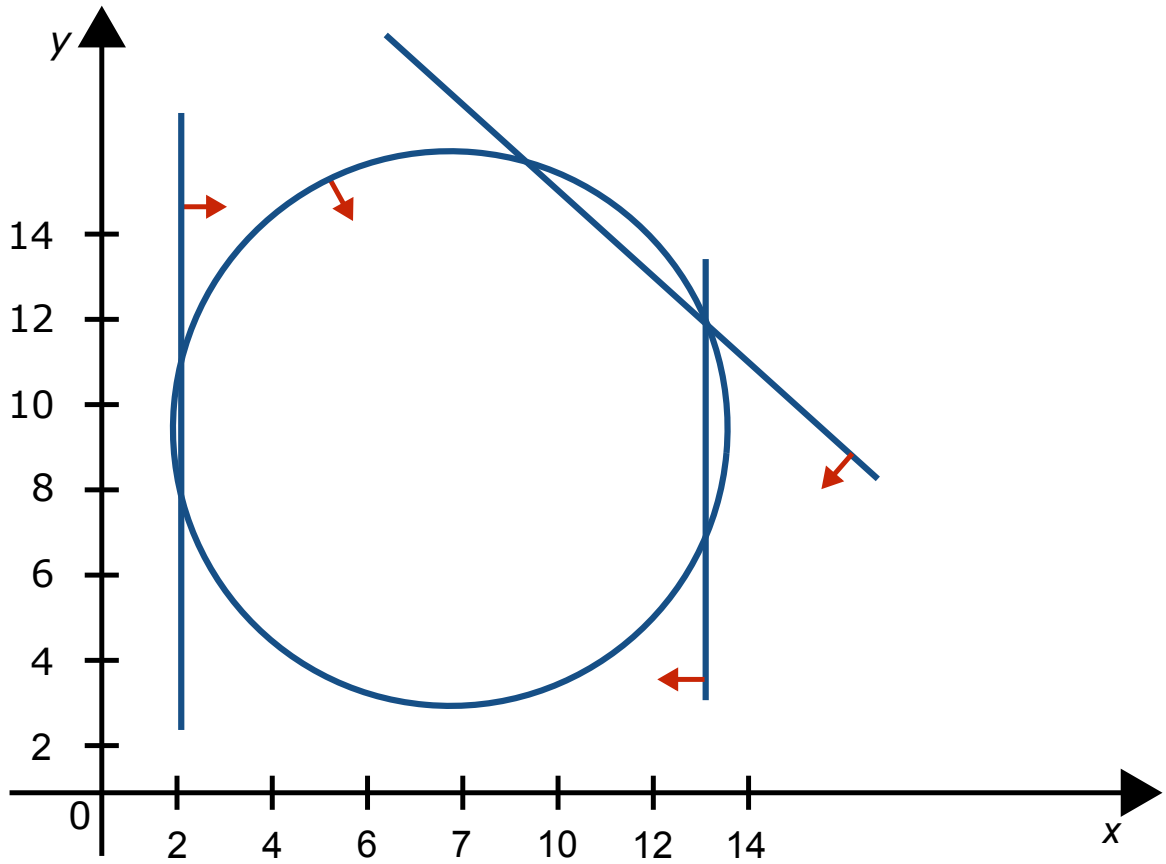
$$
\begin{aligned}
\text{minimize} \quad & (x-11)^2 + (y-10)^2 \\
\text{subject to} \quad & (x-8)^2 + (y-9)^2 \leq 49 \\
& 2 \leq x \leq 13 \\
& x + y \leq 24 \\
& y \geq 0
\end{aligned}
$$

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Solving Non-Linear Programs: Graphical Solution

Step 1: Plot each of the constraints

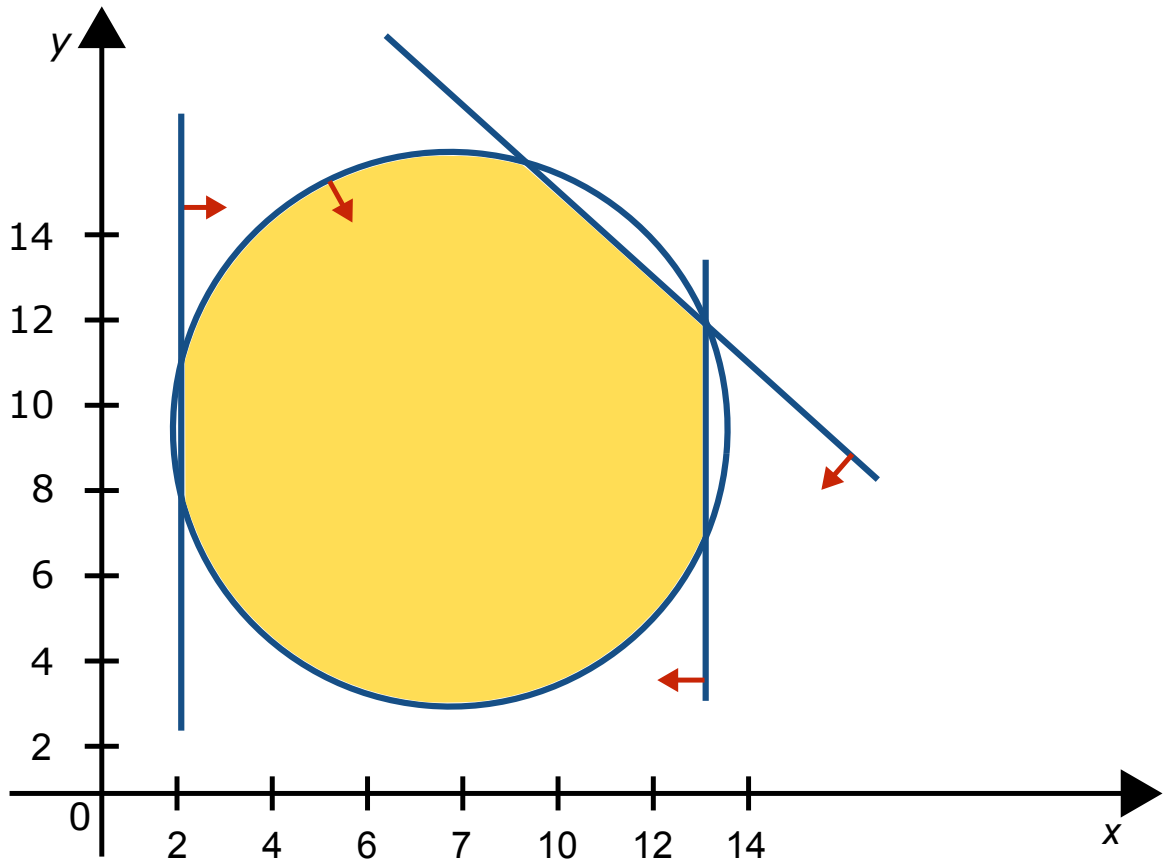
- plot the constraint equation line
- determine which of the two regions satisfies the constraint



Solving Non-Linear Programs: Graphical Solution

Step 2: Plot the feasible region

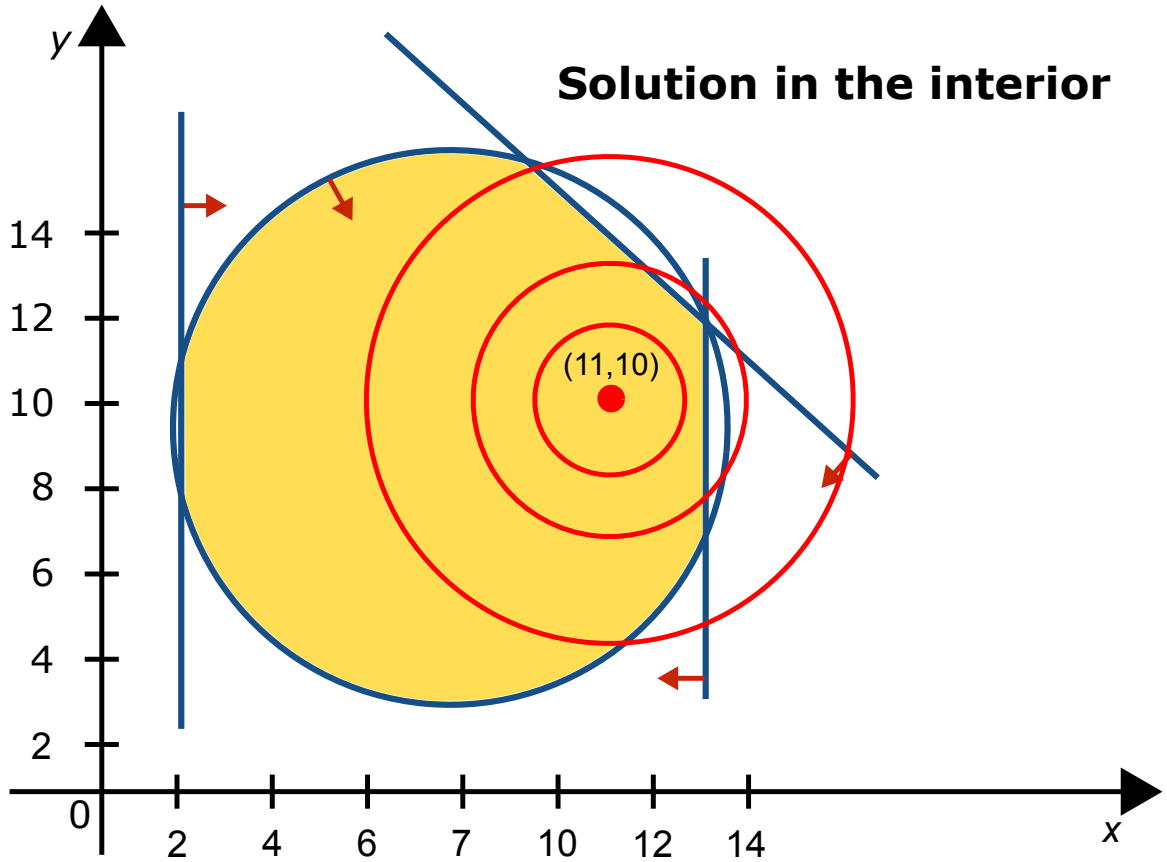
- Check which area in the plane satisfies all constraints.



Solving Non-Linear Programs: Graphical Solution

Step 3: Determine objective function isoquants

- Draw all solutions that achieve a particular, fixed objective value

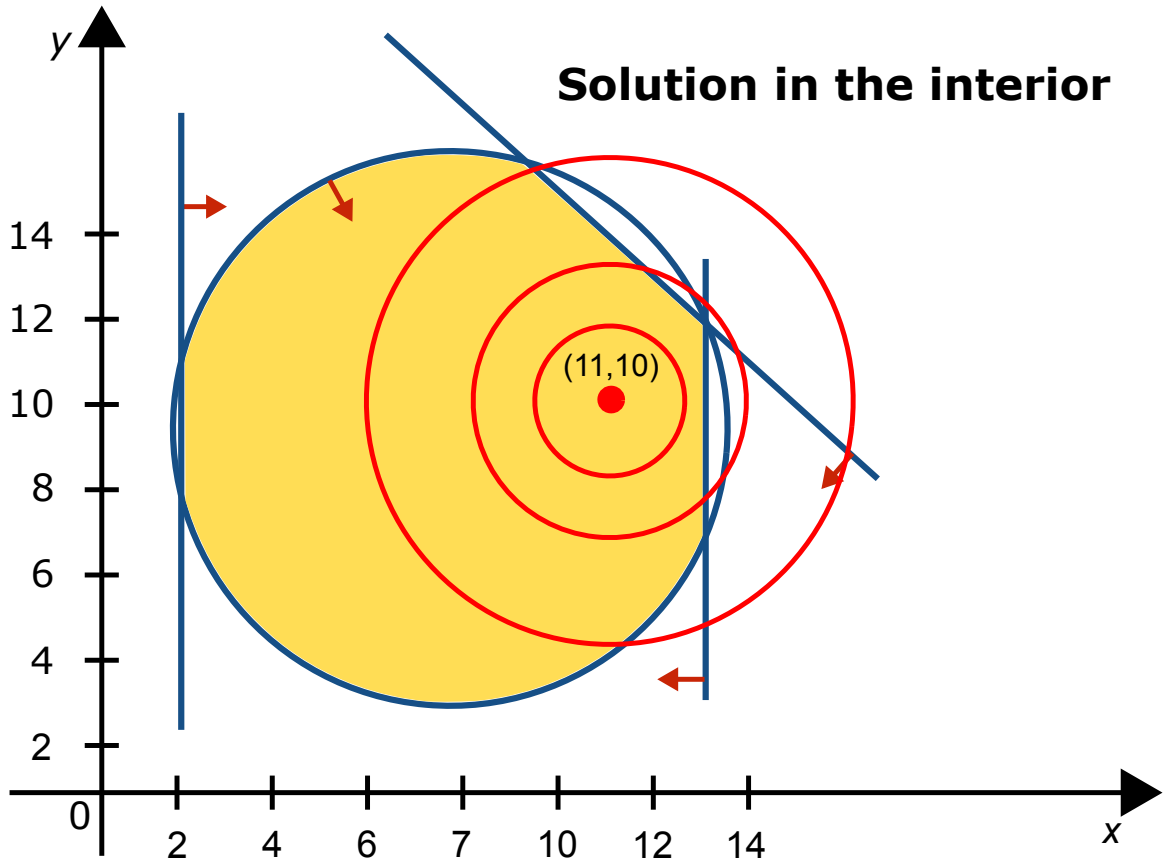


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Solving Non-Linear Programs: Graphical Solution

Step 3: Determine objective function isoquants

- Draw all solutions that achieve a particular, fixed objective value



Optimal solution: **does not occur at the boundary** of the feasible region!

Solving Non-Linear Programs: Graphical Solution

Consider the problem:

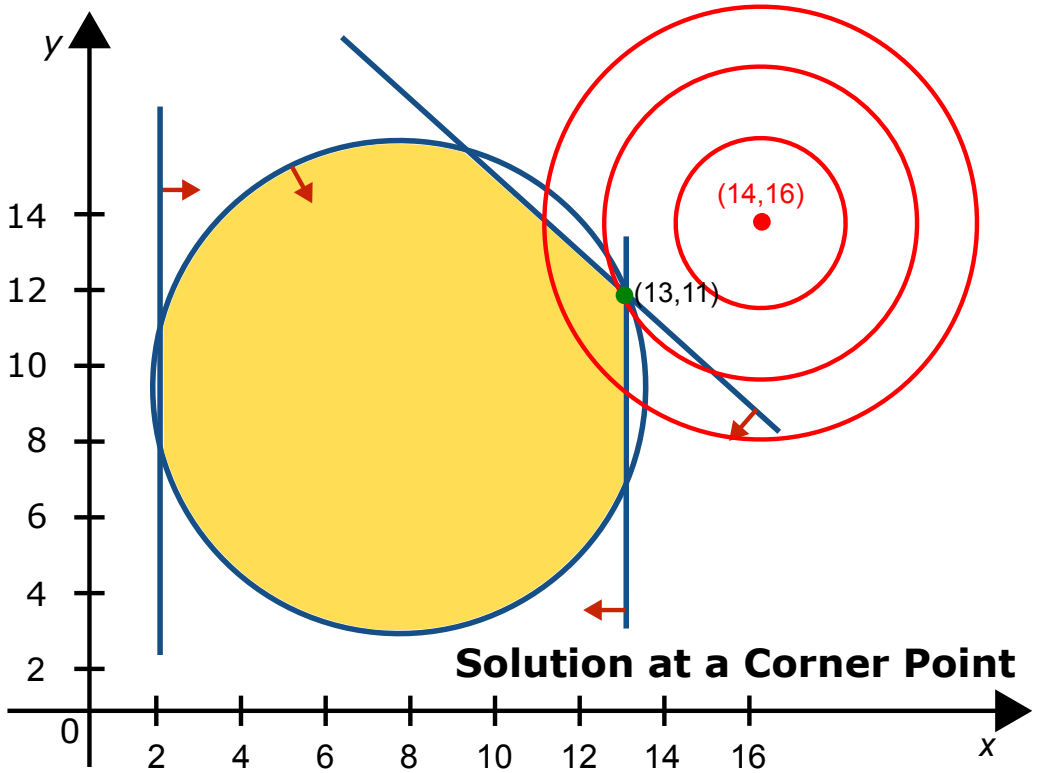
$$
\begin{aligned}
\text{minimize} \quad & (x-16)^2 + (y-14)^2 \\
\text{subject to} \quad & (x-8)^2 + (y-9)^2 \leq 49 \\
& 2 \leq x \leq 13 \\
& x + y \leq 24 \\
& y \geq 0
\end{aligned}
$$

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Solving Non-Linear Programs: Graphical Solution

Step 3: Determine objective function isoquants

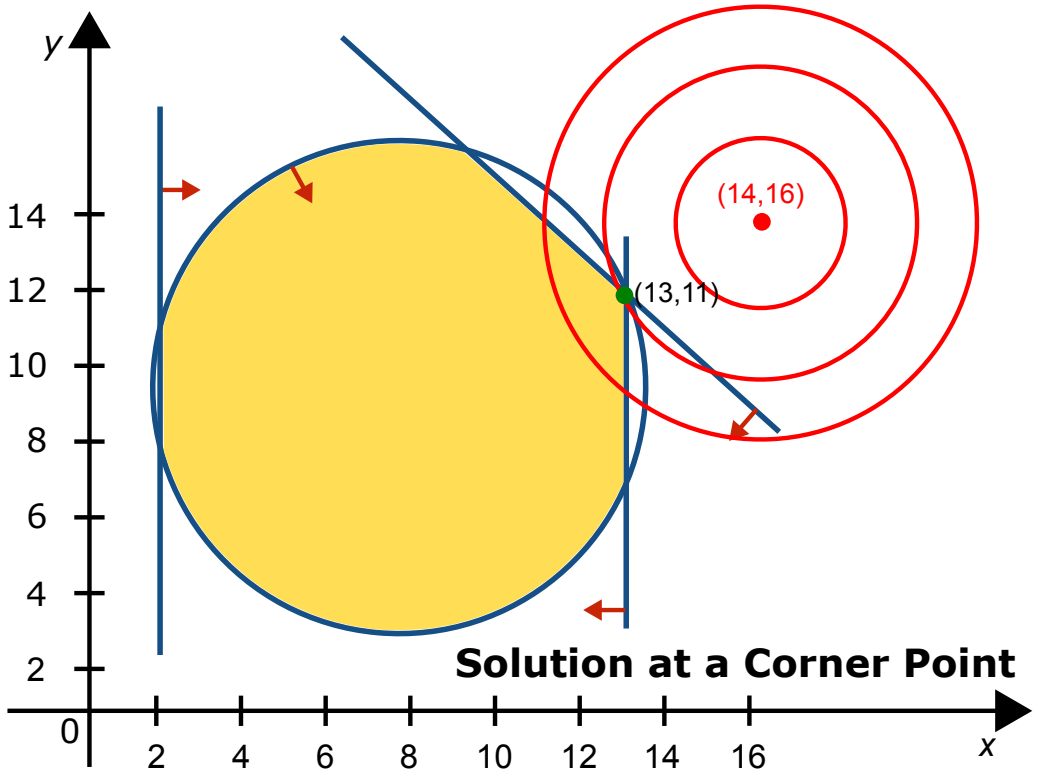
- Draw all solutions that achieve a particular, fixed objective value



Solving Non-Linear Programs: Graphical Solution

Step 3: Determine objective function isoquants

- Draw all solutions that achieve a particular, fixed objective value

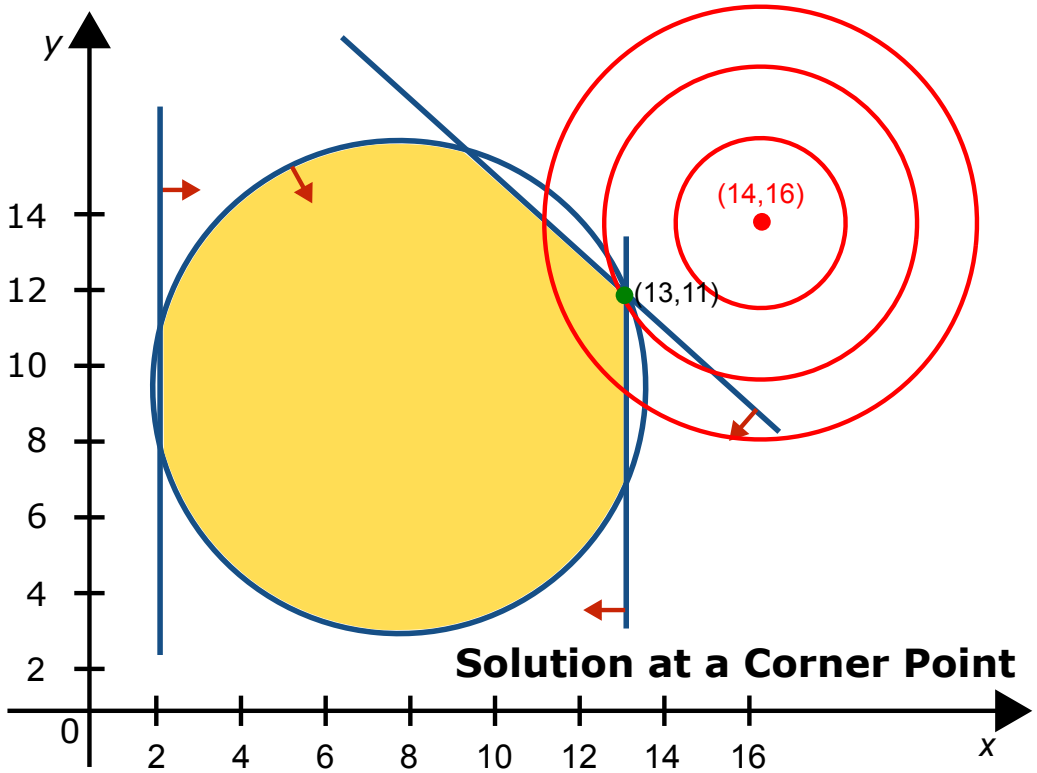


Binding Constraints: constraints $x + y \stackrel{!}{=} 24$ and $x \stackrel{!}{=} 13$ are satisfied with equality at the optimal solution.

Solving Non-Linear Programs: Graphical Solution

Step 3: Determine objective function isoquants

- Draw all solutions that achieve a particular, fixed objective value



Binding Constraints: constraints $x + y \stackrel{!}{=} 24$ and $x \stackrel{!}{=} 13$ are satisfied with equality at the optimal solution.

- **Binding Constraints** are also referred to as **Active Constraints**

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Solving Non-Linear Programs: Graphical Solution

Consider the problem:

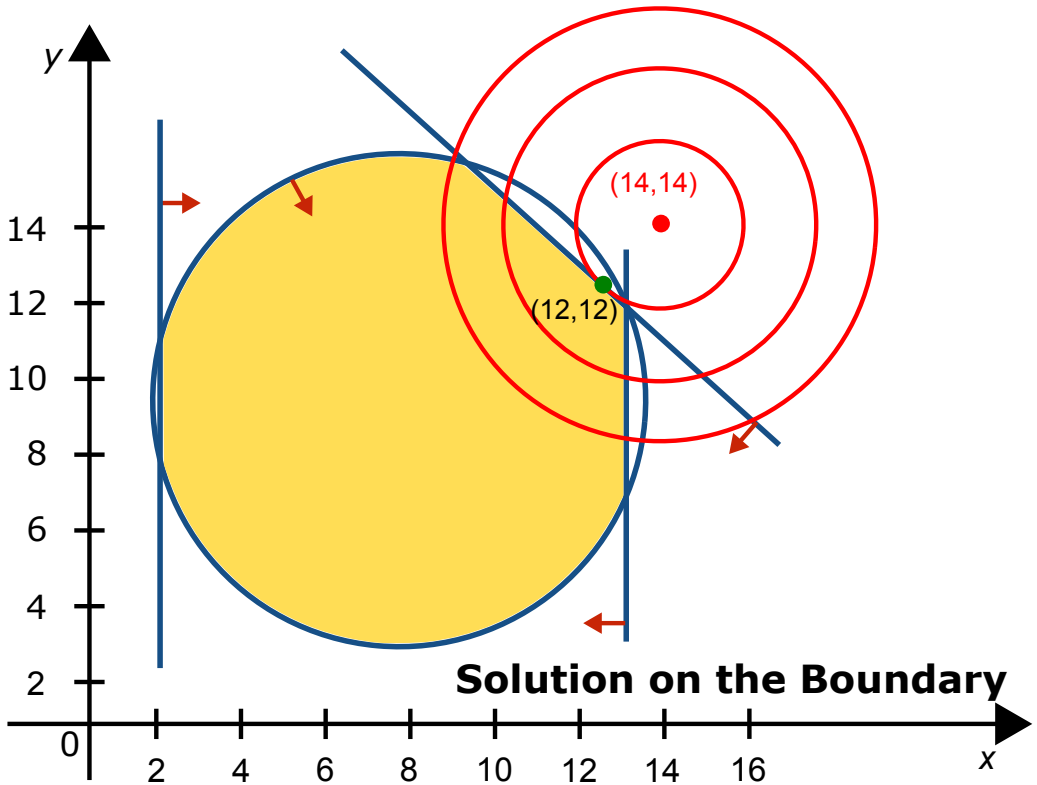
$$
\begin{aligned}
\text{minimize} \quad & (x-14)^2 + (y-14)^2 \\
\text{subject to} \quad & (x-8)^2 + (y-9)^2 \leq 49 \\
& 2 \leq x \leq 13 \\
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& y \geq 0
\end{aligned}
$$

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Solving Non-Linear Programs: Graphical Solution

Step 3: Determine objective function isoquants

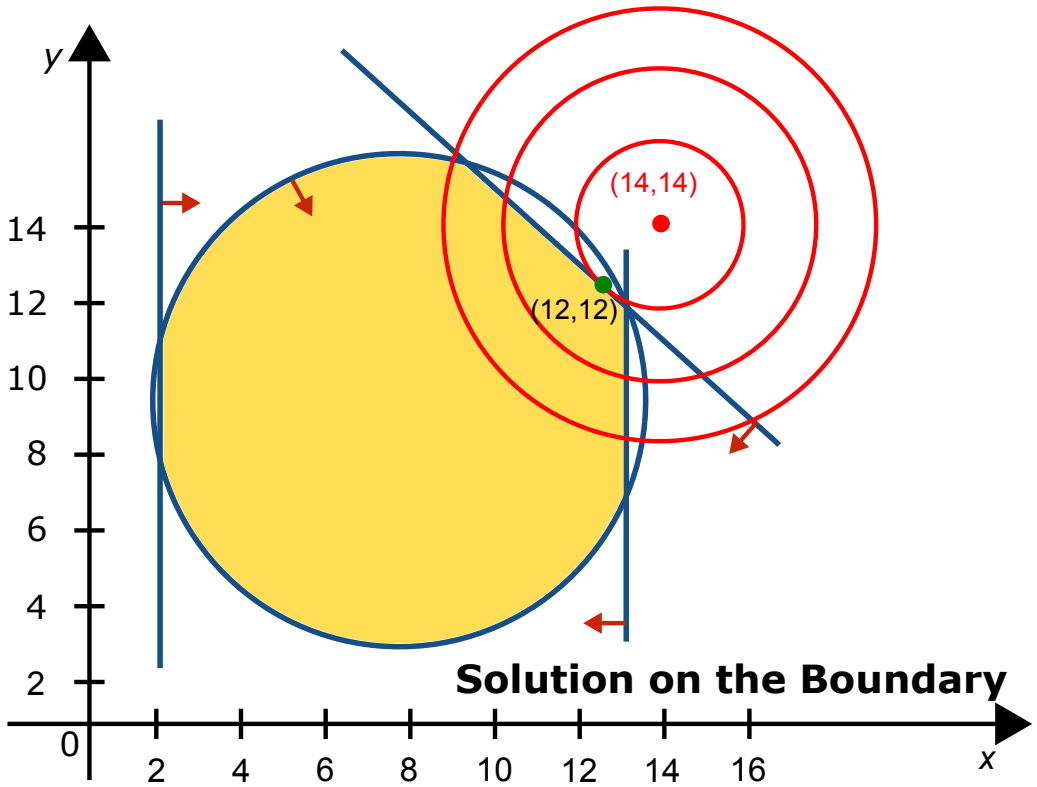
- Draw all solutions that achieve a particular, fixed objective value



Solving Non-Linear Programs: Graphical Solution

Step 3: Determine objective function isoquants

- Draw all solutions that achieve a particular, fixed objective value



Only $x + y \leq 24$ is binding!

Non-Linear Optimization Problems

In General:

- The optimal solution need **not** be at a **"corner point"**, **nor** even on the **boundary** of the feasible region.

- The **optimal solution** is where the feasible region just touches the best iso-objective value.

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Sensitivity Analysis

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Sensitivity Analysis

Consider again the problem:

$$\begin{aligned} \text{minimize} \quad & (x-14)^2 + (y-14)^2 \\ \text{subject to} \quad & (x-8)^2 + (y-9)^2 \leq 49 \\ & 2 \leq x \leq 13 \\ & x + y \leq 24 \\ & y \geq 0 \end{aligned}$$

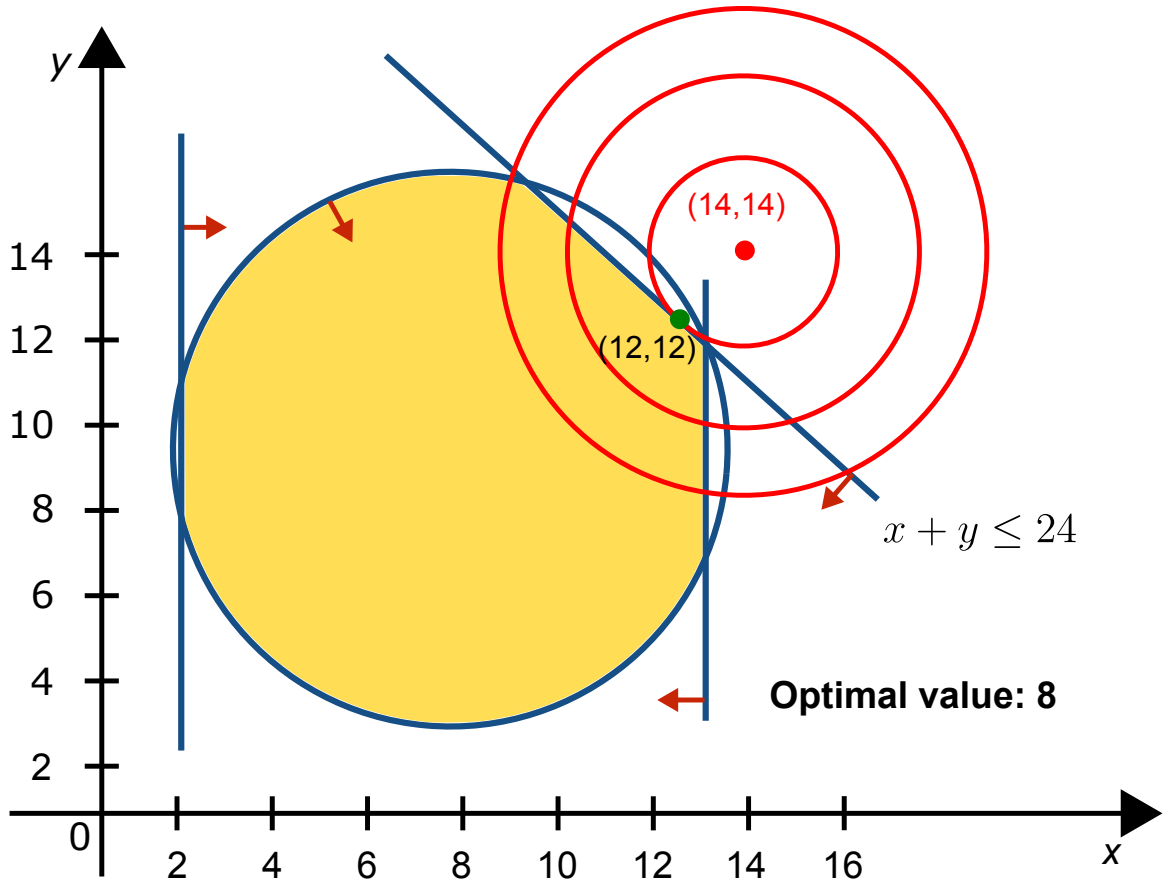
Boundary solution:

- $(x, y) = (12, 12)$
- Optimal value $= 8$

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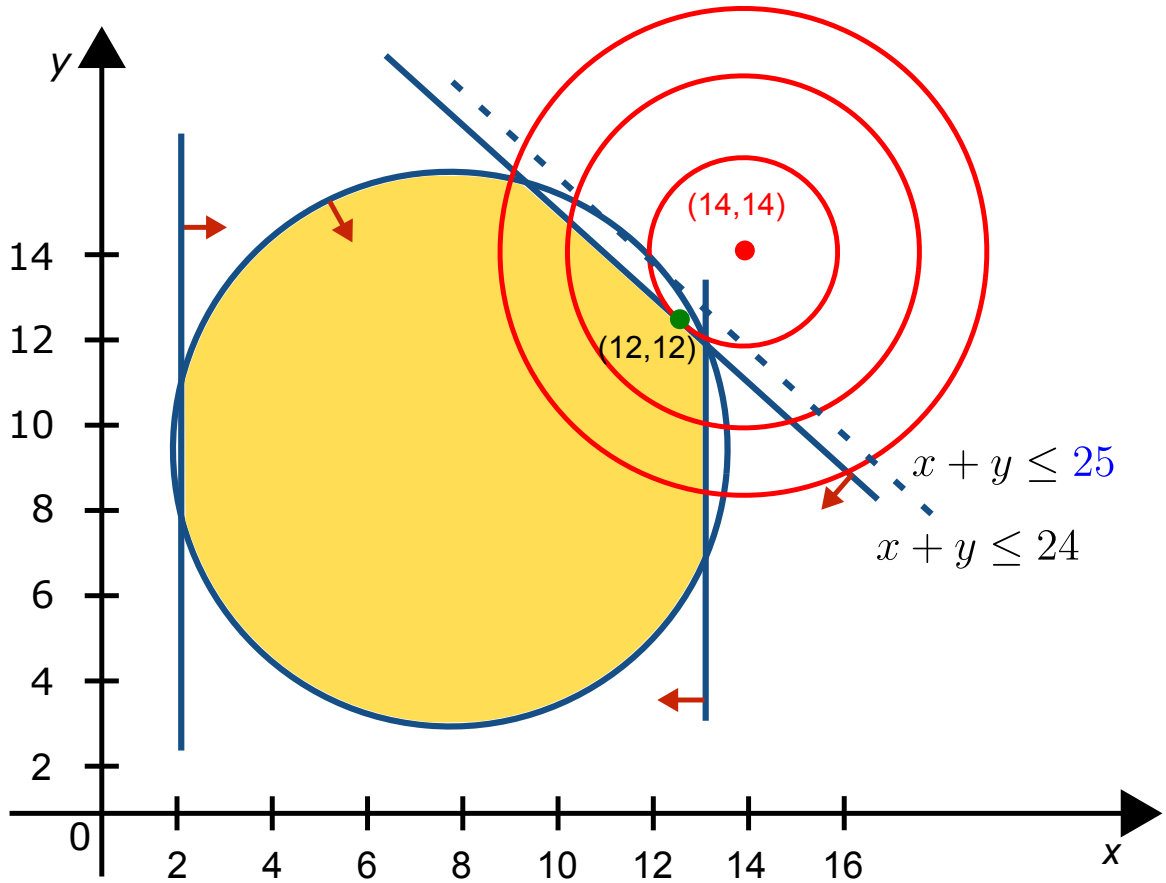
Sensitivity Analysis

Relax constraint $x + y \leq 24$ by 1 unit



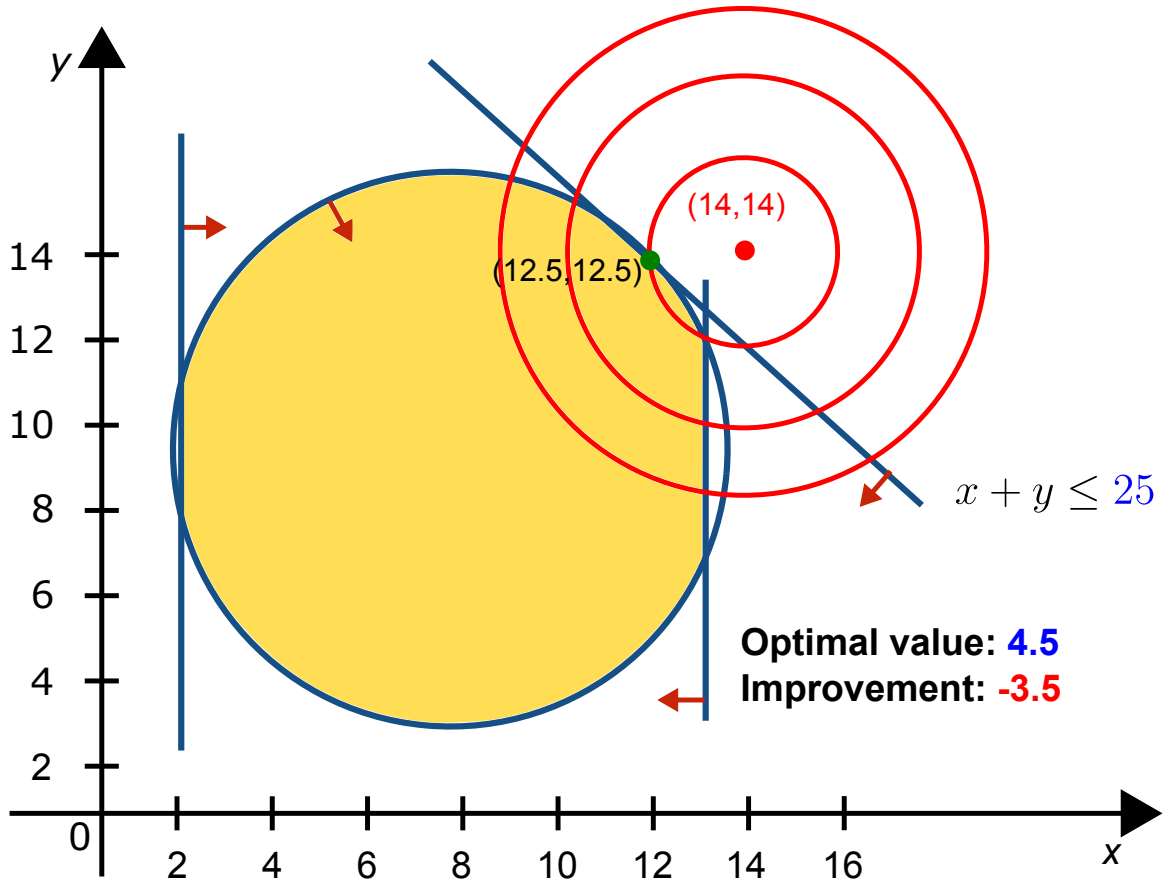
Sensitivity Analysis

Relax constraint $x + y \leq 24$ by 1 unit



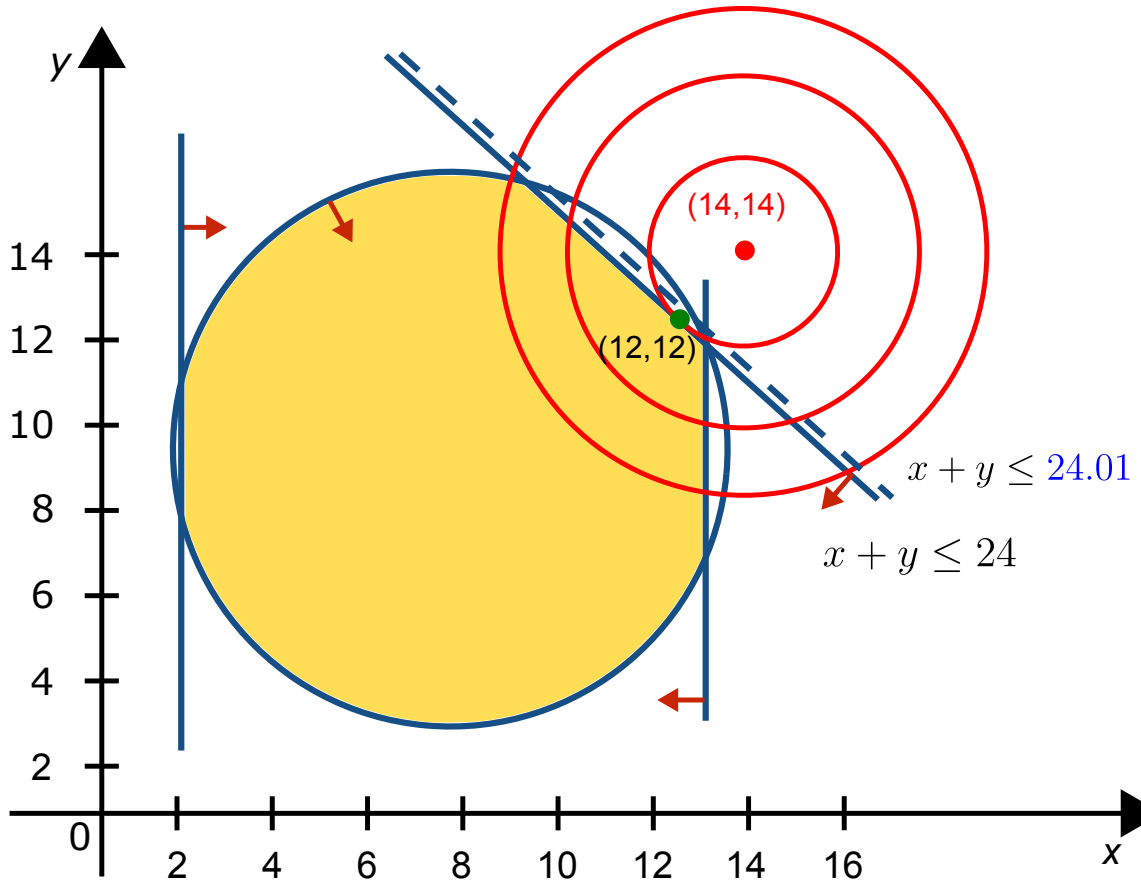
Sensitivity Analysis

Relax constraint $x + y \leq 24$ by 1 unit



Sensitivity Analysis

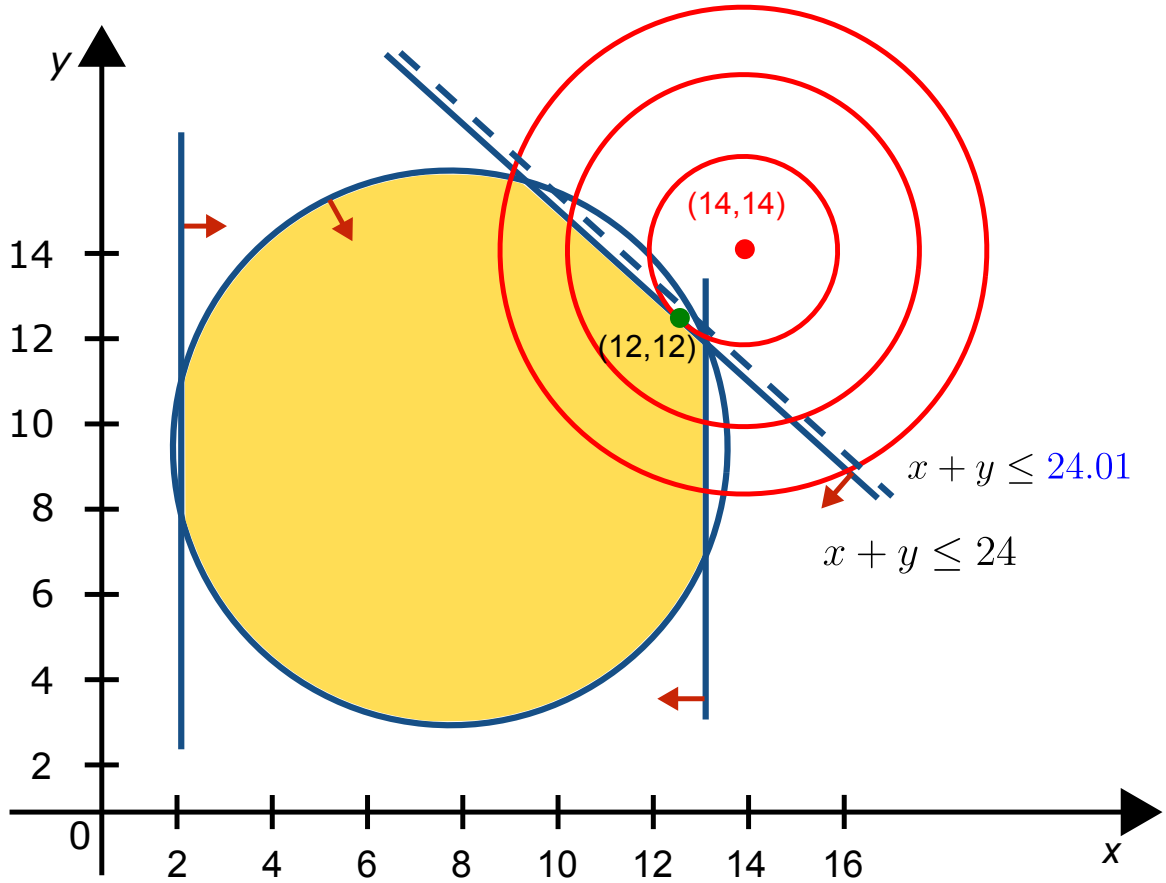
Relax constraint $x + y \leq 24$ by 1 unit



Careful: Unlike Linear Optimization, there is no range for which the shadow prices remain valid.

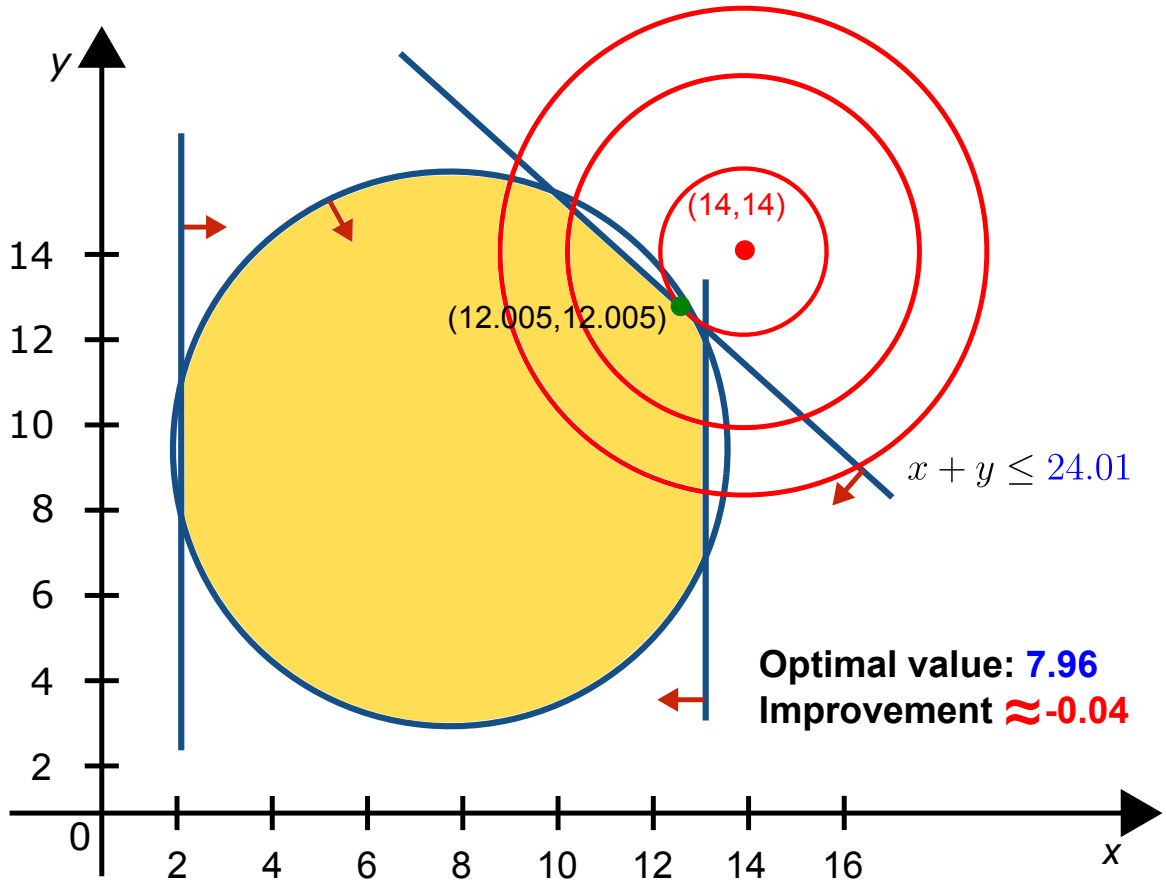
Sensitivity Analysis

However if we relax constraint $x + y \leq 24$ by 0.01 unit



Sensitivity Analysis

However if we relax constraint $x + y \leq 24$ by 0.01 unit



Shadow price:

$$\frac{\text{Improvement}}{\text{change RHS}} \approx \frac{-0.04}{0.01} = -4$$

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Sensitivity Analysis

Shadow Price Interpretation:

Optimal objective decreases at a rate of 4 units per unit of increase in the RHS of the $x + y \leq 24$ constraint.

$$\text{RHS:} \quad x + y \leq 24 \rightarrow 24 + \Delta$$

$$\text{Optimal value of problem} \approx 8 - 4\Delta$$

For Nonlinear functions is only an approximation

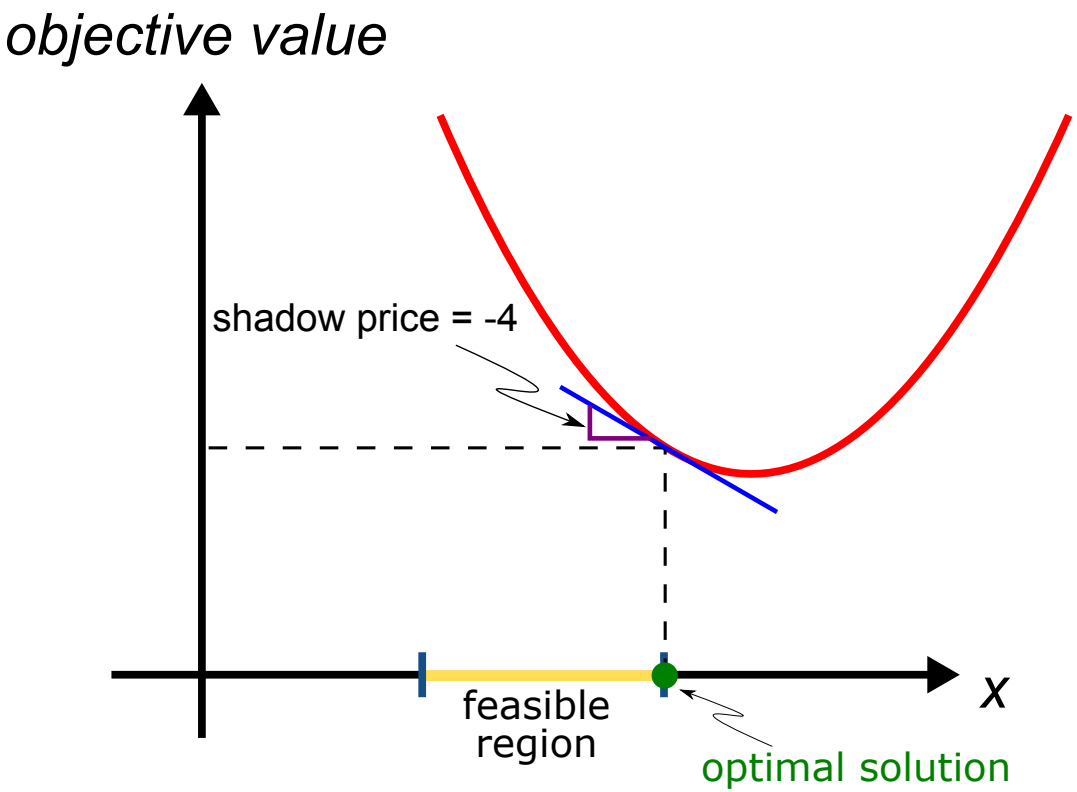
- **only valid for small Δ**

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Sensitivity Analysis

Why?

- The shadow price is **the tangent** of the change in the optimal objective relative to a change in the RHS of the constraint.
- It is only a point estimate **(only gives local information)**.



If constraint is **non-binding** at optimal solution, then its **shadow price is zero**.

Gradient Descent Method

for Solving

Unconstrained Nonlinear Programs

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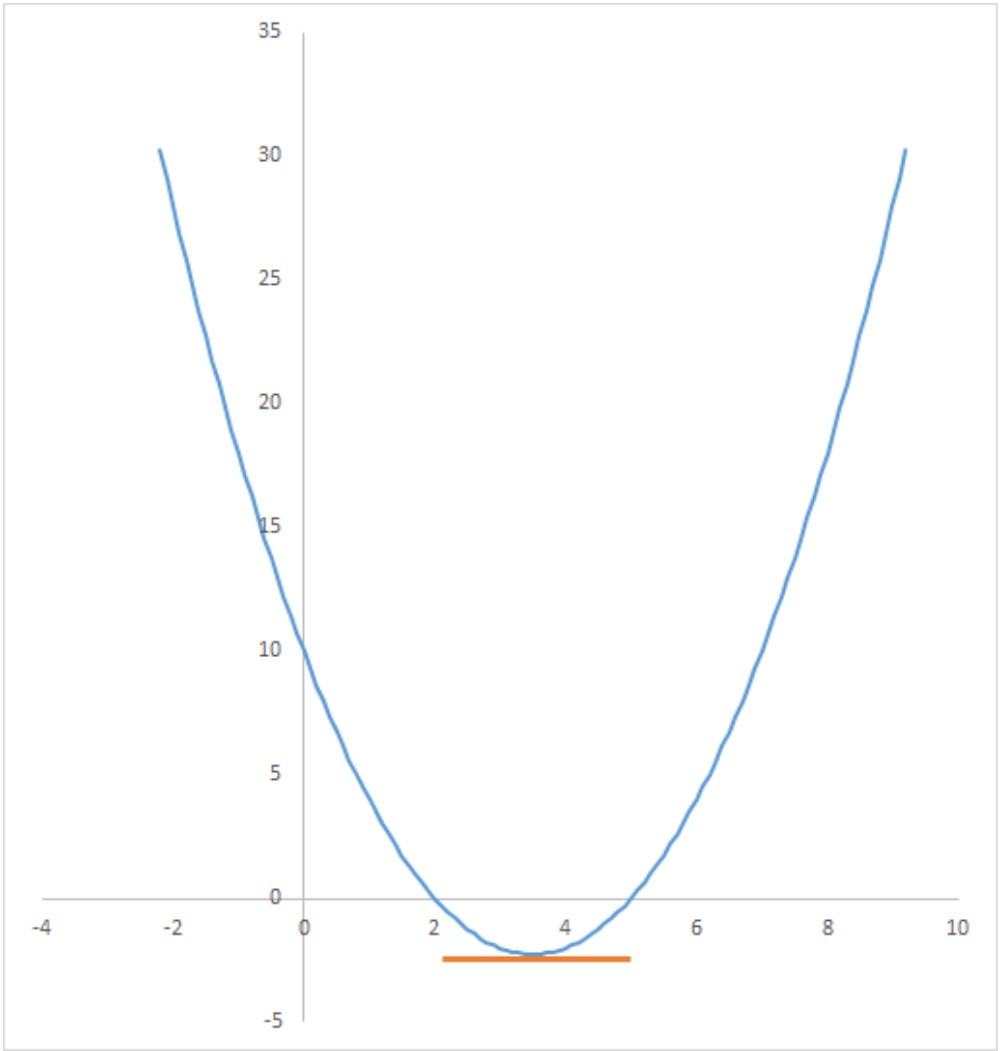
Unconstrained Optimization

Assume we want to $\min_x f(x) = x^2 - 7x + 10$ with $x \in \mathbb{R}$. A graph of the function is below:

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Unconstrained Optimization

Assume we want to $\min_x f(x) = x^2 - 7x + 10$ with $x \in \mathbb{R}$. A graph of the function is below:



How shall we go about finding the minimum point?
Differentiate! Take the derivative and equate to 0:

$$\begin{aligned} \frac{df(x)}{dx} &= 2x - 7 \\ &= 0 \end{aligned}$$

from which we conclude $x = \frac{7}{2}$ and $f(\frac{7}{2}) = -2.25$. Simple!

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How shall we go about finding the minimum point?
Differentiate! Take the derivative and equate to 0:

$$\begin{aligned} \frac{df(x)}{dx} &= 2x - 7 \\ &= 0 \end{aligned}$$

from which we conclude $x = \frac{7}{2}$ and $f(\frac{7}{2}) = -2.25$. Simple!

Why was this so simple?

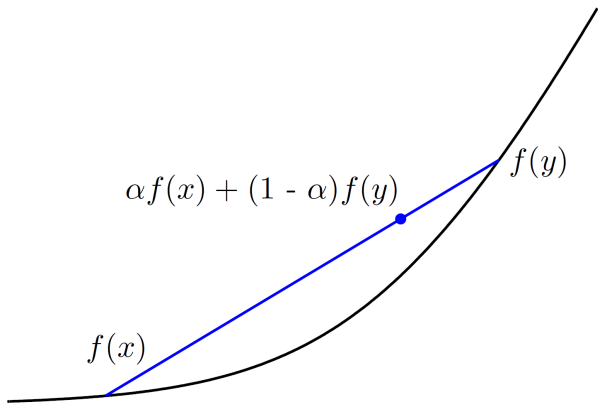
- The function behaves well: it is convex.
- The feasible region has a nice form: it is convex.

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What Is a Convex Function?

A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if for x and y in the domain of f and $\alpha \in [0, 1]$,

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y).$$



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What If Non-Convex?

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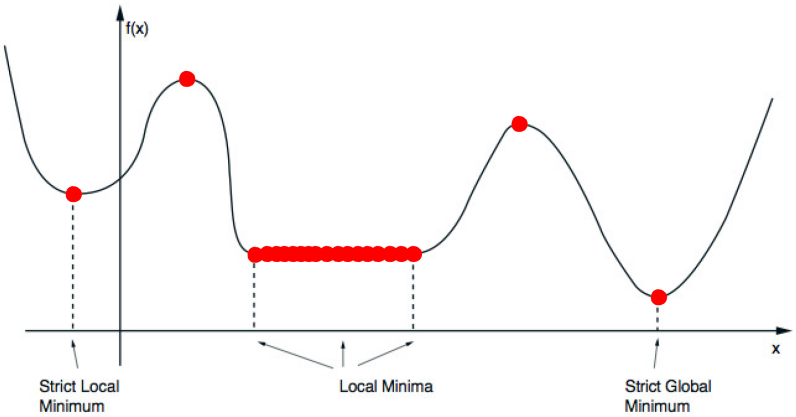
What If Non-Convex?

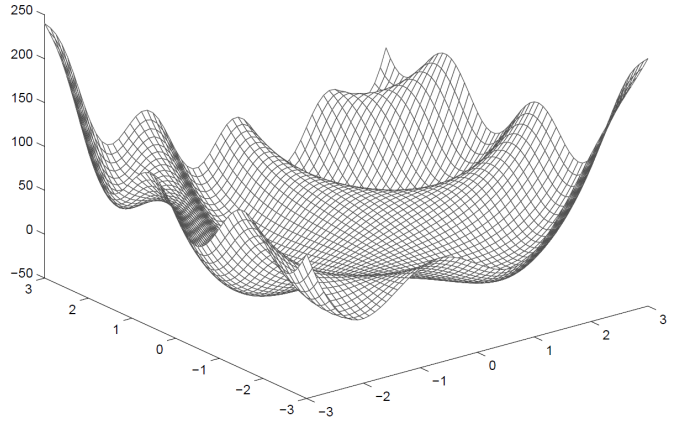
Then we have trouble!

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What If Non-Convex?

Then we have trouble! Look at the functions below:





Remark: There are things to do to help (a bit) but out of scope of this course.

But We Are on Safe Ground with Convex Functions, Right?

Yes! Sort of.

- The good news is for a convex function, if you find a local optimum x^* (at which the derivative is zero), then you have also found the global optimum.

- The bad news is you need to compute the derivative and also solve the system of equations that result from equating the derivative to 0 which is easier said than done.

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Gradient Descent for Convex, Unconstrained Minimization Problems

Gradient descent (for minimization problems) requires to compute the derivative but not to solve the system of equations that results from equating the derivative to 0.

1. Take a point in the domain of $f(x)$ such as x_0 and compute $f(x_0)$.
2. If $\Delta f = f(x_0^{\text{New}}) - f(x_0^{\text{Old}}) \leq \epsilon$, stop. x_0 is the optimal solution and $f(x_0)$ is the optimal value.
3. Otherwise, compute the derivative of $f(x)$ at x_0.
4. Set $x_0 \leftarrow x_0 - \eta \frac{\partial f}{\partial x}(x_0)$.
5. Go back to Step 2.

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Gradient Descent: An Example

Consider $f(x) = x^2 - 7x + 10$. Let's see how this method works: (Assume $\eta = \frac{1}{7}$).

Iteration 1: Take $x_0 = 0$; so $f(0) = 10$ and $\frac{df}{dx}(x = 0) = -7$.

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Iteration 2: $x_0 \leftarrow 0 - \frac{1}{7}(-7) = 1$; so $f(1) = 4$ and $\frac{df}{dx}(x = 1) = -5$.

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Iteration 3: $x_0 \leftarrow 1 - \frac{1}{7}(-5) = \frac{12}{7}$; so $f(\frac{12}{7}) = \frac{46}{49}$ and $\frac{df}{dx}(x = \frac{12}{7}) = -\frac{25}{7}$.

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Gradient Descent: An Example

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Iteration 4: . . . (See file).

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Gradient Descent: Another Example

Consider $f(x, y) = (x - 5)^2 + (y - 8)^2$. Apply the gradient descent method to minimize the function.

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Gradient Descent: Another Example

Consider $f(x, y) = (x - 5)^2 + (y - 8)^2$. Apply the gradient descent method to minimize the function. (See file).

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Gradient Descent: In-class Exercise

Eight data points (x, y) represent the price of a house (in 1000 dollars) as a function of the number of bedrooms are given as

House	Price y	Bedrooms x
A	\$90	1
B	\$120	1
C	\$150	1
D	\$150	2
E	\$200	2
F	\$180	4
G	\$250	4
H	\$400	5

Formulate an NLP to find the coefficients of the best regression line to predict the price as a function of the number of bedrooms. Draw the function and apply the gradient descent method to find the optimal coefficients.

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Gradient Descent: In-class Exercise

For a regression line $\hat{y} = \beta_0 + \beta_1 x_i$, we have $RSS = \sum_i (y_i - (\beta_0 + \beta_1 x_i))^2$. so:

$$\begin{aligned} \frac{\partial RSS}{\partial \beta_0} &= -2 \sum_i (y_i - (\beta_0 + \beta_1 x_i)) \\ \frac{\partial RSS}{\partial \beta_1} &= -2 \sum_i (y_i - (\beta_0 + \beta_1 x_i)) x_i \end{aligned}$$

Assume $\eta = 0.01$, and $\epsilon = 0.001$

- **Step 1** Start at $(\beta_0, \beta_1) = (0, 0)$. Compute RSS.
- **Step 2** Compute the gradient at (β_0, β_1).
- **Step 3** Set $(\beta_0, \beta_1) \leftarrow (\beta_0 - \eta \frac{dRSS}{d\beta_0}, \beta_1 - \eta \frac{dRSS}{d\beta_1})$.
- **Step 4** Is $\Delta RSS < \epsilon$, stop. If not, return to **Step 2**.

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Non-Linear Optimization Problems

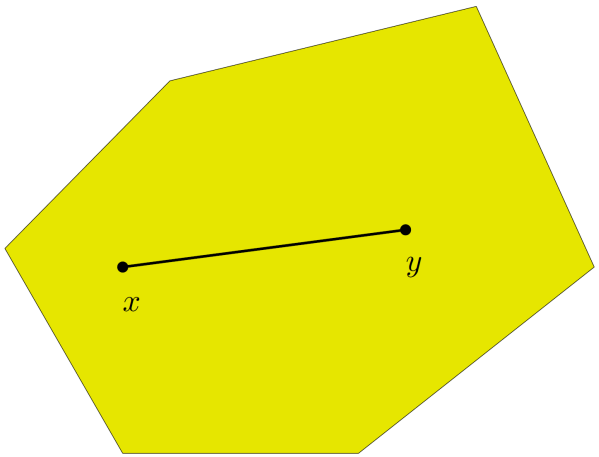
with Constraints

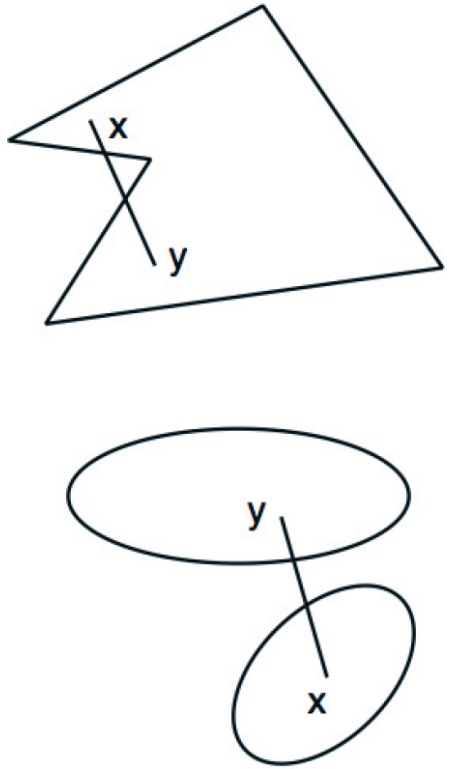
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What Is a Convex Feasible Region?

A set $C \subseteq \mathbb{R}^n$ is convex if for x and y in C and $\alpha \in [0, 1]$,

$$\alpha x + (1 - \alpha)y \in C.$$





Nonconvex Sets

Convex Constrained Optimization Problems

A nonlinear optimization problem is called a convex minimization problem if

1. The objective function is a convex function to be minimized.
2. Equality constraints are linear.
3. Constraints of the form $\leq$ are convex.
4. (For constraints of the form $\geq$, multiply by a -1 and check (3).)

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How To Solve Convex Constrained Optimization Problems?

- **Good News:** Many machine learning algorithms (such as linear regression and logistic regression) are unconstrained convex optimization problems provided that one is smart in the choice of the function to optimize.
  - For instance, in logistic regression we do not use RSS. It won't be convex and get us in trouble; I mean big trouble!

- **Bad News:** Gradient descent method does not work with constrained optimization problems.

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How to Solve Convex Constrained Optimization Problems?

Let's see an example:

$$\begin{aligned} \text{maximize} \quad & f(x, y) = xy \\ \text{subject to} \quad & x + y^2 \leq 2 \\ & x, y \geq 0 \end{aligned}$$

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We can write this problem as:

$$\begin{aligned} \text{maximize} \quad & xy \\ \text{subject to} \quad & x + y^2 \leq 2 \\ & -x \leq 0 \\ & -y \leq 0 \end{aligned}$$

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How to Solve Convex Constrained Optimization Problems?

Interpret the following function:

$$L(x, y, \lambda_1, \lambda_2, \lambda_3) = xy - \lambda_1(x + y^2 - 2) - \lambda_2(-x - 0) - \lambda_3(-y - 0).$$

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How to Solve Convex Constrained Optimization Problems?

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Let's bring our malevolent market player back...

- If $x + y^2 > 2$, the market will set λ_1 a very high positive number. This will penalize L greatly. So we must have $x + y^2 \leq 2$.
- If $x + y^2 < 2$, then the market will set $\lambda_1 = 0$.

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- So at optimality, either $x + y^2 < 2$ and $\lambda_1 = 0$ or $x + y^2 = 0$ and $\lambda_1 \geq 0$. Note that λ_1 is the penalty and it will never be negative.
- Overall, we have $\lambda_1(x + y^2 - 2) = 0$. We can likewise argue that $\lambda_2(-x) = 0$ and $\lambda_3(-y) = 0$. We called these

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- Overall, we have $\lambda_1(x + y^2 - 2) = 0$. We can likewise argue that $\lambda_2(-x) = 0$ and $\lambda_3(-y) = 0$. We called these "complementary slackness conditions".

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How to Solve Convex Constrained Optimization Problems?

We then do with function L what we do with other unconstrained optimization problems: take derivatives and set them equal to 0.

$$\begin{aligned} \frac{\partial L}{\partial x} &= y - \lambda_1 + \lambda_2 = 0 \\ \frac{\partial L}{\partial y} &= x - 2y\lambda_1 + \lambda_3 = 0 \end{aligned}$$

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Karush-Kuhn-Tucker Conditions for Constrained Optimization

Let's put all these together:

$$\begin{aligned} y - \lambda_1 + \lambda_2 &= 0 \\ x - 2y\lambda_1 + \lambda_3 &= 0 \\ \lambda_1(x + y^2 - 2) &= 0 \\ \lambda_2(-x) &= 0 \\ \lambda_3(-y) &= 0 \\ x + y^2 &\leq 2 \\ x, y, \lambda_1, \lambda_2, \lambda_3 &\geq 0. \end{aligned}$$

These are called Karush-Kuhn-Tucker conditions (or equations) for optimality.

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Karush-Kuhn-Tucker Conditions for Constrained Optimization

Case 1: Suppose $\lambda_1 = 0$. Then the first KKT condition says $y + \lambda_2 = 0$ and the second says $x + \lambda_3 = 0$. Since each term is nonnegative, the only way that can happen is if $x = y = \lambda_2 = \lambda_3 = 0$. Indeed, the KKT conditions are satisfied when $x = y = \lambda_1 = \lambda_2 = \lambda_3 = 0$.

But clearly this is not a local maximum since $f(0, 0) = 0$ while $f(x, y) > 0$ at points in the interior of the feasible region.

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Karush-Kuhn-Tucker Conditions for Constrained Optimization

Case 2: Suppose $x + y^2 = 2$. Now are least one of $x = 2 - y^2$ and y must be positive.

- **Case 2a:** Suppose $x > 0$. Then $\lambda_2 = 0$. The first KKT condition says $x - 2y\lambda_1 + \lambda_3 = 0$, so $3y^2 = 2 + \lambda_3 > 0$ and $\lambda_3 = 0$. Thus $y = \sqrt{2/3}$, and $x = 2 - 2/3 = 4/3$. Again all KKT conditions are satisfied.

- **Case 2b:** Suppose $x = 0$, i.e., $y = \sqrt{2}$. Since $y > 0$, we have $\lambda_3 = 0$. From the second KKT condition, we must have $\lambda_1 = 0$. But this takes us back to Case 1.

We conclude that there are only two candidates for a local max: $(0, 0)$ and $(4/3, \sqrt{2/3})$. The global maximum is at $(4/3, \sqrt{2/3})$.

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KKT Conditions for NL Constrained Optimization Problems: In-class Exercise

$$\begin{aligned} \text{minimize} \quad & (x-2)^2 + 2(y-1)^2 \\ \text{subject to} \quad & x + 4y \leq 3 \\ & x \geq y \end{aligned}$$

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KKT Conditions for Optimality

Step 1: Turn the problem into the standard form:

$$\begin{aligned} \text{maximize} \quad & -(x-2)^2 - 2(y-1)^2 \\ \text{subject to} \quad & x + 4y - 3 \leq 0 \\ & -x + y \leq 0 \end{aligned}$$

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KKT Conditions for Optimality

Step 1: Turn the problem into the standard form:

$$\begin{aligned} \text{maximize} \quad & -(x-2)^2 - 2(y-1)^2 \\ \text{subject to} \quad & x + 4y - 3 \leq 0 \\ & -x + y \leq 0 \end{aligned}$$

Step 2: Write the Lagrangian

$$L(x, y, \lambda_1, \lambda_2) = -(x-2)^2 - 2(y-1)^2 - \lambda_1(x + 4y - 3) - \lambda_2(-x + y)$$

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KKT Conditions for Optimality

Step 3: Write the KKT conditions:

$$
\begin{aligned}
-2(x-2) - \lambda_1 + \lambda_2 &= 0 \\
-4(y-1) - 4\lambda_1 - \lambda_2 &= 0 \\
\lambda_1(x + 4y - 3) &= 0 \\
\lambda_2(-x + y) &= 0 \\
x + 4y - 3 &\leq 0 \\
-x + y &\leq 0 \\
\lambda_1, \lambda_2 &\geq 0.
\end{aligned}
$$

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KKT Conditions for Optimality

Step 4: Solve the system of equations:

1. Assume $\lambda_1 = \lambda_2 = 0$. So it must be that $x = 2$ and $y = 1$; but this is not admissible because $x + 4y - 3 \nleqslant 0$.
2. Assume $\lambda_1 = 0$ and $\lambda_2 > 0$. It follows that $x = 4/3$ and $y = 4/3$; but this is not admissible because $x + 4y - 3 \nleqslant 0$.
3. Assume $\lambda_2 = 0$ and $x + 4y - 3 = 0$. Therefore, $x = 5/3$, $y = 1/3$, and $\lambda_1 = 2/3$.
4. Assume $x = y$ and $x + 5y - 3 = 0$. It follows that $x = y = 3/5$, $\lambda_1 = 22/25$, and $\lambda_2 = -48/25$ which is not admissible because λ_2 must be nonnegative.

So the optimal solution is $x = 5/3$, $y = 1/3$, and $f(x, y) = -4/9$.

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Practical Solution to KKT Conditions

- For practical problems with many DVs and constraints, solving the KKT equations is difficult.
- So instead techniques such as "sequential quadratic programming" (SQP) is applied to obtain solutions.
- SQP in nature is similar to gradient descent; we won't cover this method in this class.

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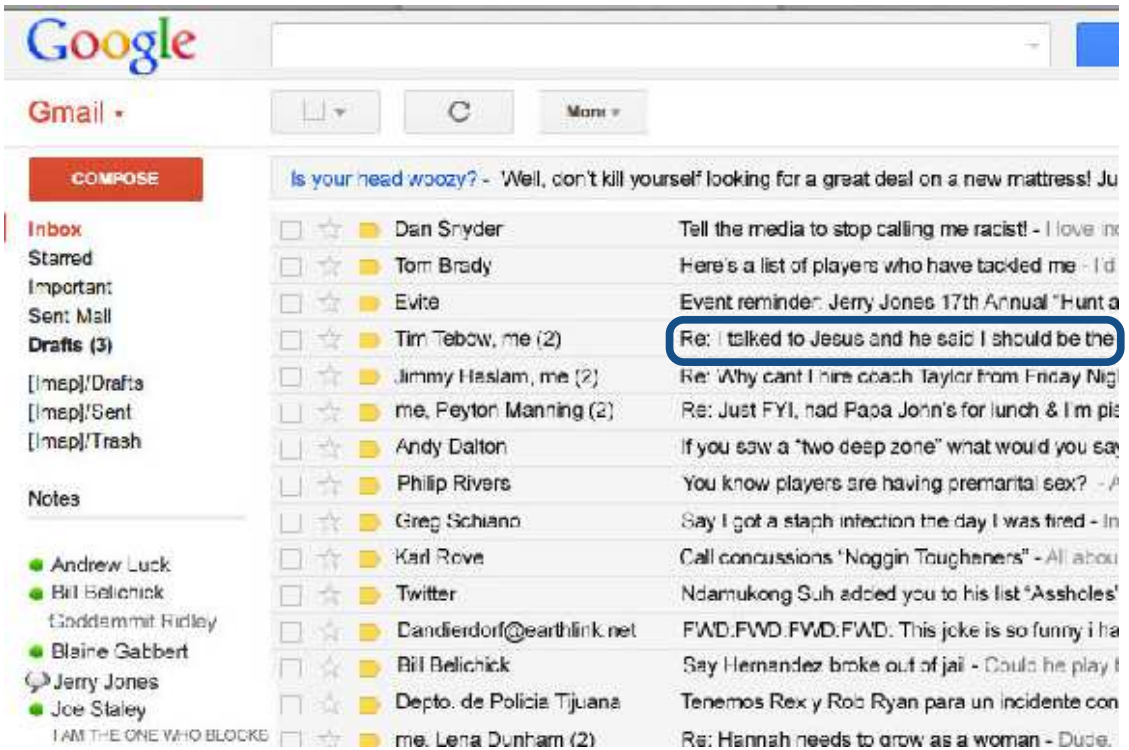
Other Applications of NLP

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Support Vector Machines

Pattern classification problem: Given a set of data points with characteristics that are classified in two ways (possess a property or not) and a new data point, does the new point possess the property?

- **Detecting spam email:**



data points: past emails

- **characteristics:** words contained in subject titles

property: spam or not?

Support Vector Machines

Pattern classification problem: Given a set of data points with characteristics that are classified in two ways (possess a property or not) and a new data point, does the new point possess the property?

- **Judging the risk for a heart attack:**



data points: medical reports

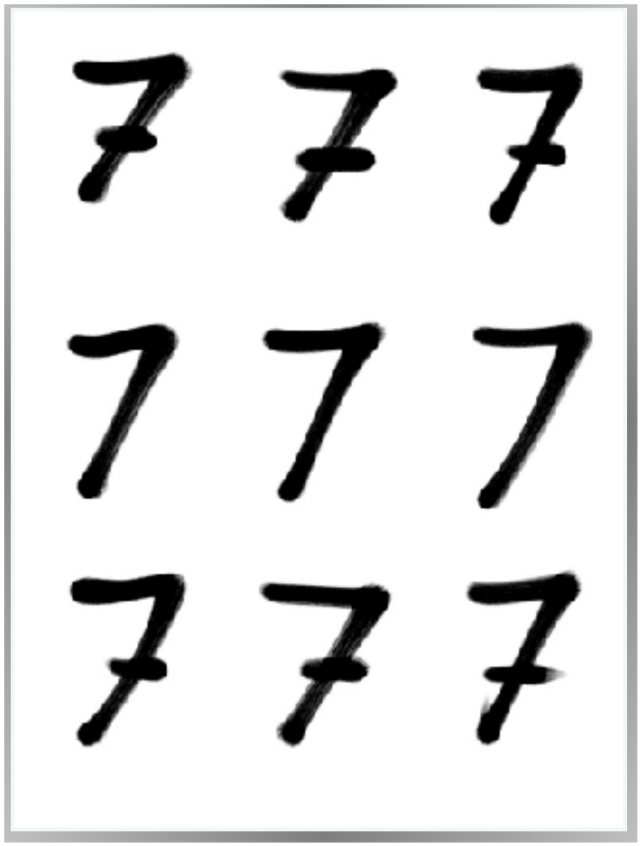
- **characteristics:** smoker?, healthy lifestyle?, weight, blood pressure, ...

property: at risk of heart attack or not?

Support Vector Machines

Pattern classification problem: Given a set of data points with characteristics that are classified in two ways (possess a property or not) and a new data point, does the new point possess the property?

- **Text recognition:**



data points: samples of handwritten digits

- **characteristics:** ratio height:width, curvature, number of strokes, …

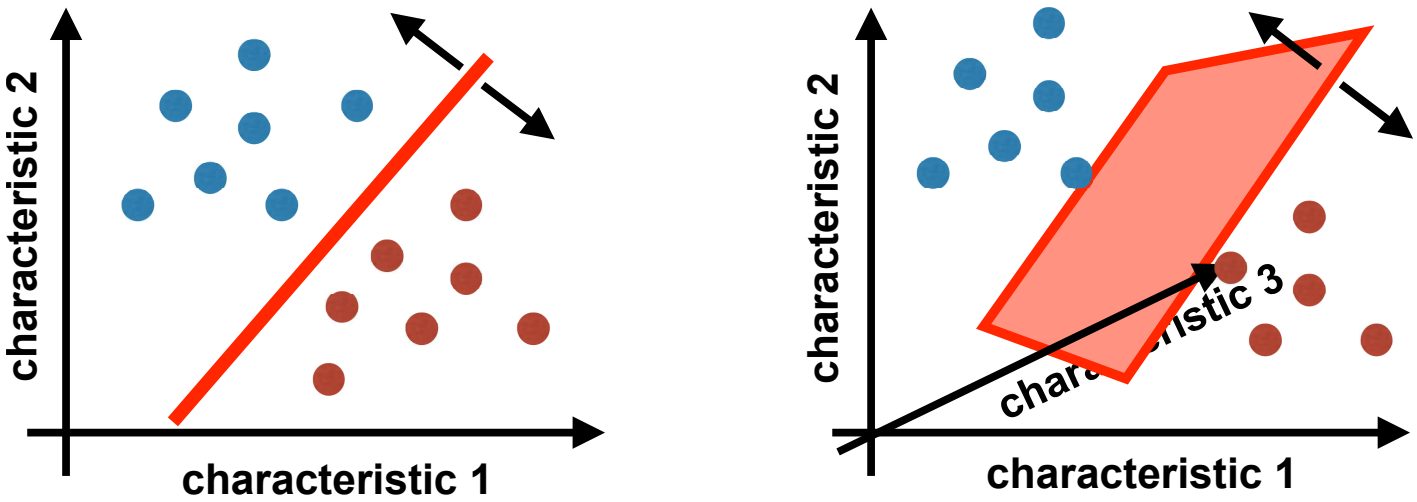
property: is it a particular digit or not?

Support Vector Machines

Pattern classification problem: Given a set of data points with characteristics that are classified in two ways (possess a property or not) and a new data point, does the new point possess the property?

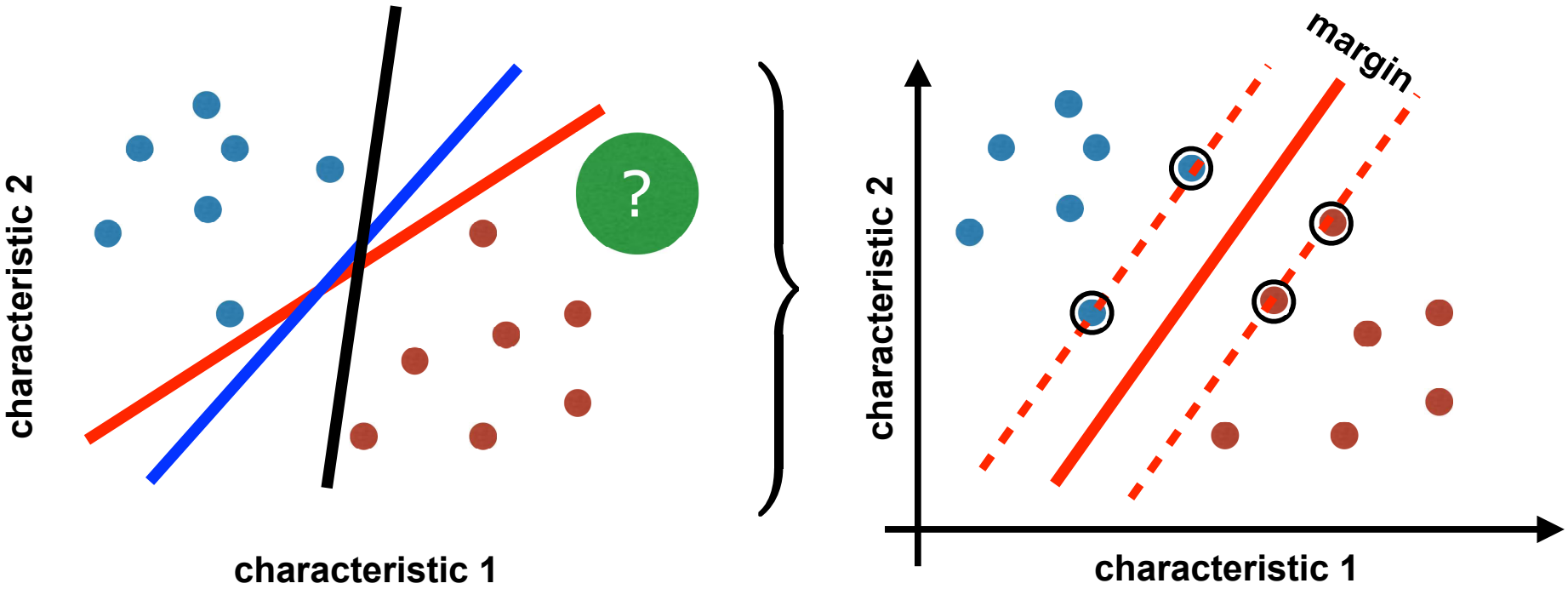
Idea: Identify a **rule** based on the existing data points (the training data) that characterises the set of data points having the property

We focus on linear rules here:



Support Vector Machines

Which linear rule should we select?

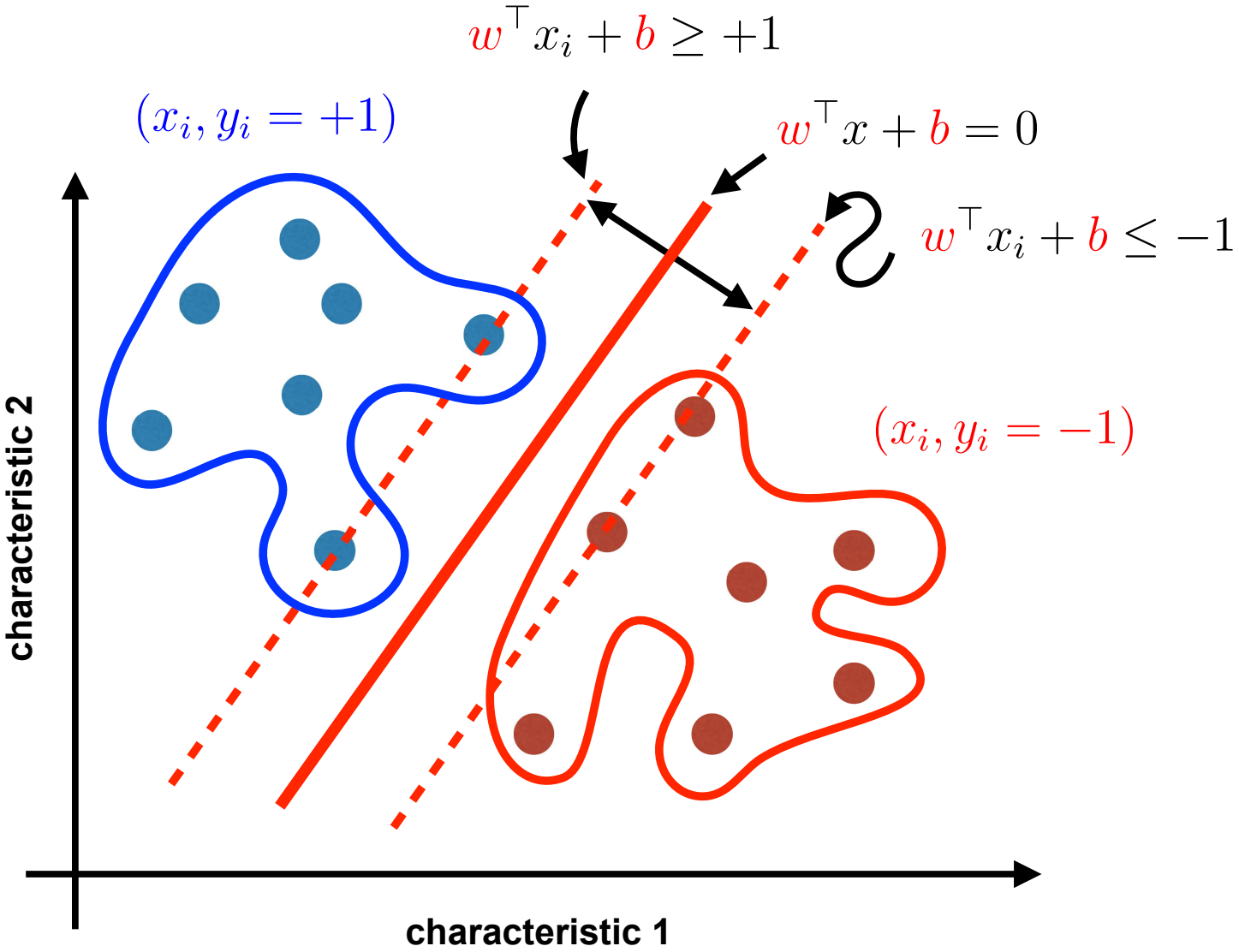


the circled data points determine the margin and are called ***support vectors***

select the rule with the ***largest margin*** between the two classes

Support Vector Machines: The Separable Case

Constraint: find a strict classifier



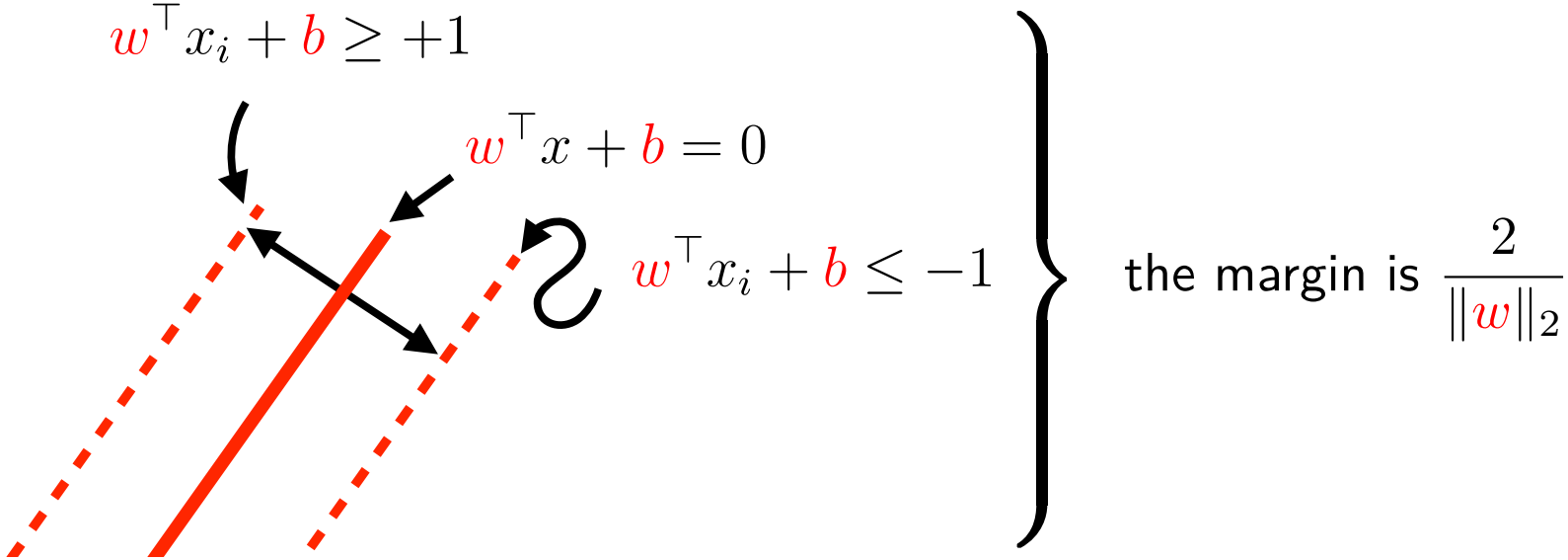
Support Vector Machines: The Separable Case

Objective: Maximize margin

The **distance between two parallel hyperplanes**

$$w^\top x + b_1 = 0 \quad \text{and} \quad w^\top x + b_2 = 0$$

is $\dfrac{|b_1 - b_2|}{\|w\|_2}$



Support Vector Machines: The Separable Case

Optimization Problem:

$$\begin{array}{lll} \text{maximize} & \dfrac{2}{\|w\|_2} & \\ \text{subject to} & w^\top x_i + b \geq +1 & \forall i : y_i = +1 \\ & w^\top x_i + b \leq -1 & \forall i : y_i = -1 \end{array}$$

Equivalent Problem:

$$\begin{array}{lll} \text{minimize} & w^\top w & \\ \text{subject to} & w^\top x_i + b \geq +1 & \forall i : y_i = +1 \\ & w^\top x_i + b \leq -1 & \forall i : y_i = -1 \end{array}$$

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Support Vector Machines: Iris identification

Automatic identification: Identifying variations of the iris flower

Iris-versicolor



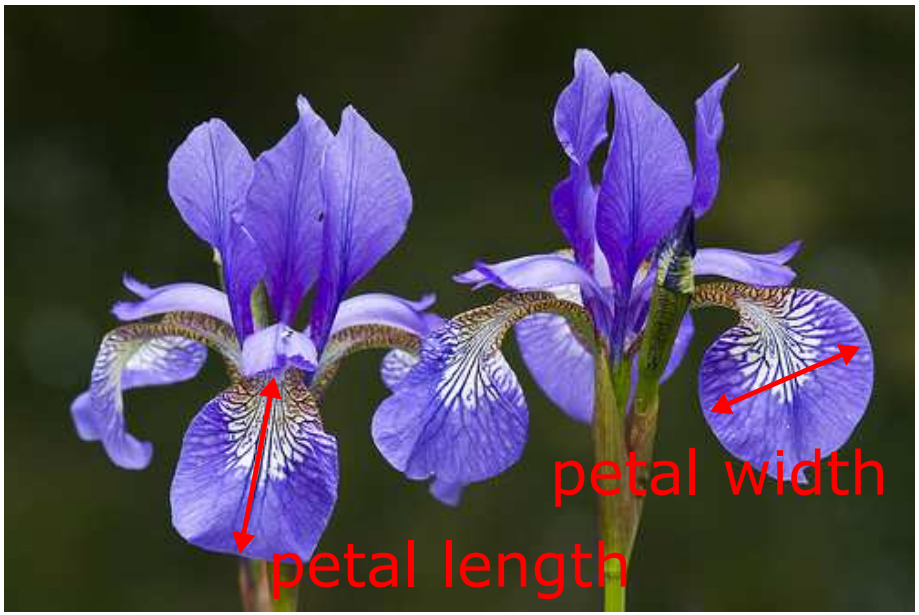
Iris-setosa



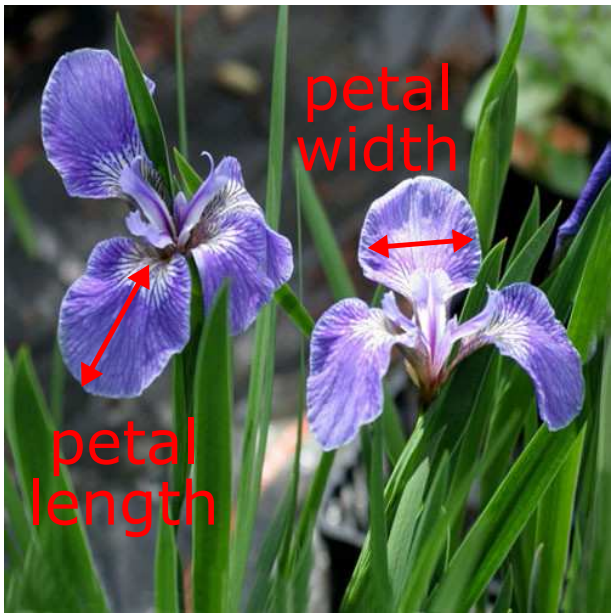
Support Vector Machines: Iris identification

Automatic identification: Identifying variations of the iris flower

Iris-versicolor

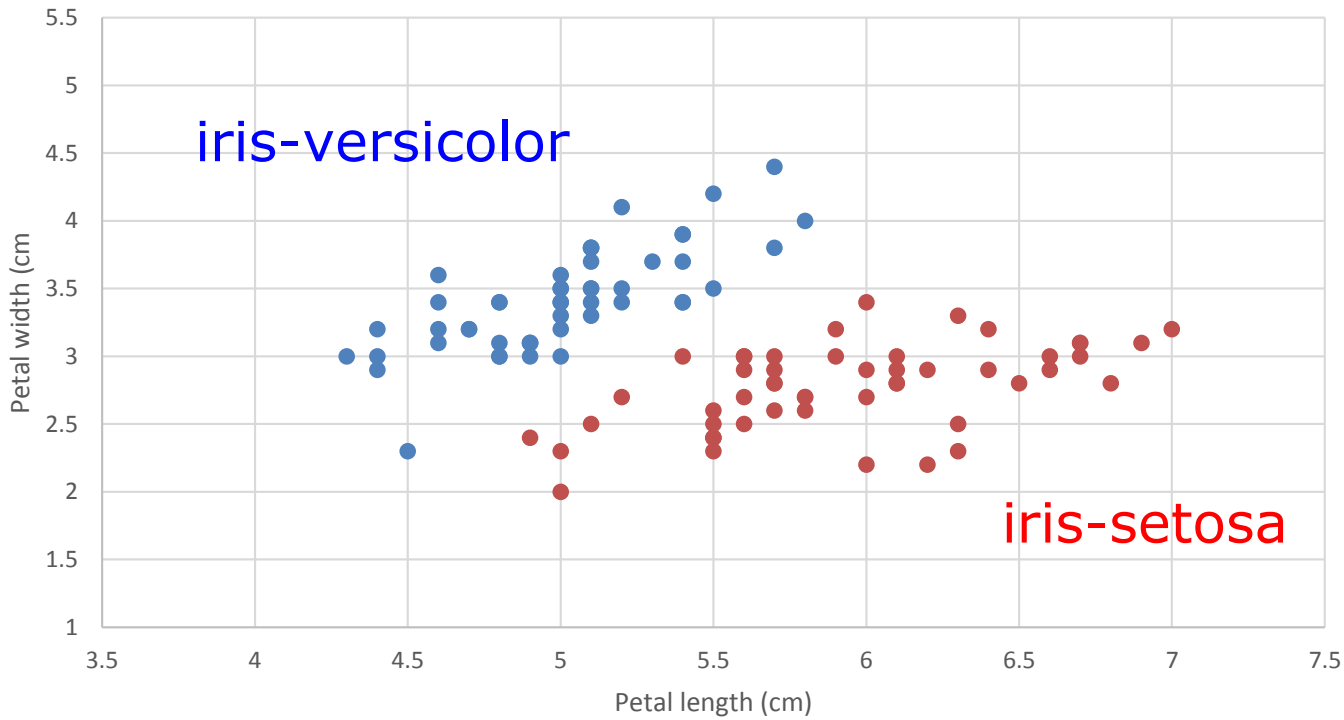


Iris-setosa



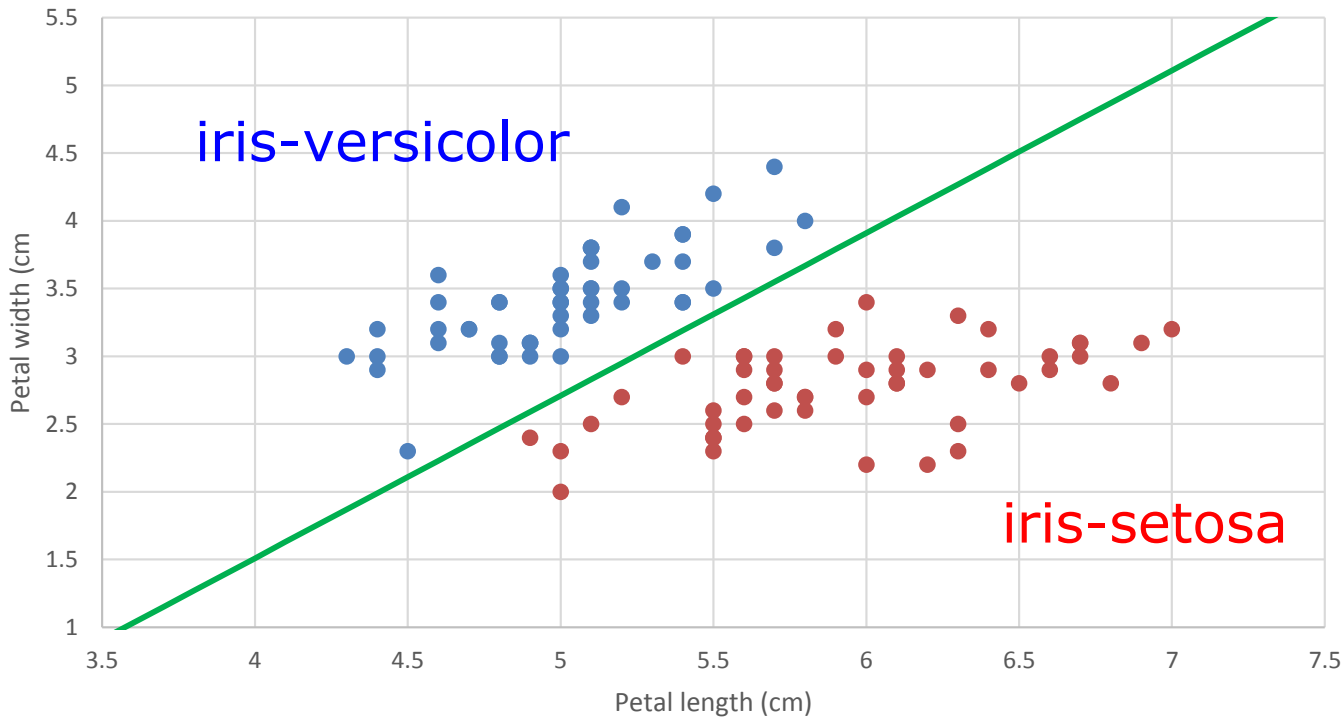
Support Vector Machines: Iris identification

Automatic identification: Identifying variations of the iris flower



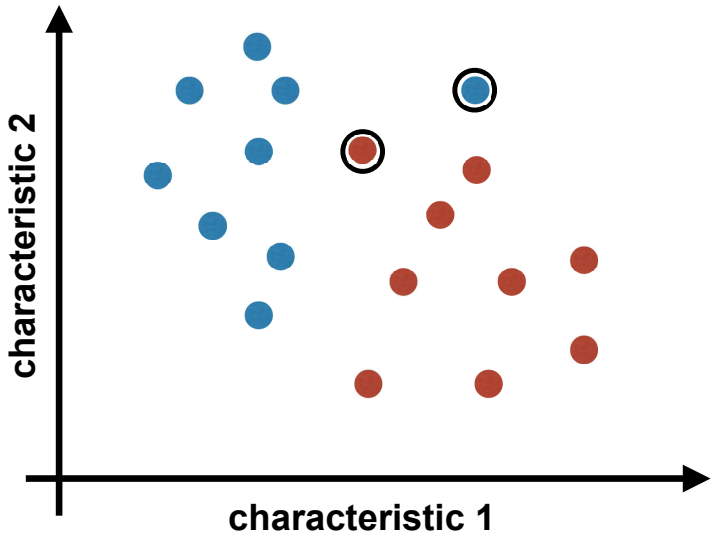
Support Vector Machines: Iris identification

Automatic identification: Identifying variations of the iris flower



Support Vector Machines: The Non-separable Case

Consider the following instance:



There is *no linear rule* that correctly classifies all data points!

Remedies:

- nonlinear rules
- "soft margin" rules

Optimization Problem:

$$\begin{aligned} \text{minimize} \quad & w^\top w + C \sum_i \xi_i \\ \text{subject to} \quad & w^\top x_i + b \geq +1 - \xi_i \qquad \forall i : y_i = +1 \\ & w^\top x_i + b \leq +1 + \xi_i \qquad \forall i : y_i = -1 \end{aligned}$$

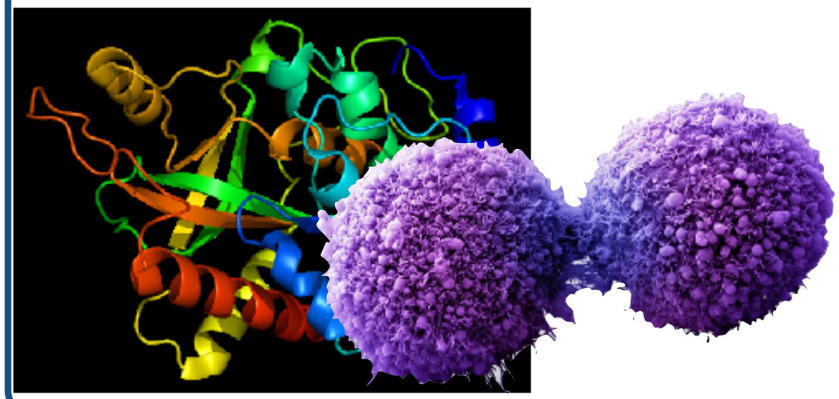
violations ξ_i are penalized in the objective

Support Vector Machines: (Some of the) Applications

Text categorisation



Bioinformatics



Recognising handwriting

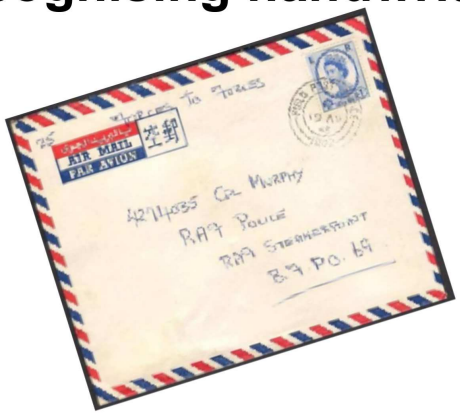


Image classification



Real-Life Applications and Software

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Nonlinear Optimization: Real-Life Applications

Bank Hapoalim Group:

- Israel's largest banking group
- 327 branches in Israel
- 37 worldwide branches

A major part of the business involves providing investment advises. An OR team was formed to do this.



In one recent year, the bank's investment advisors held

- 133,000 consultations (63,000 customers) using this decision-support system.

The annual earnings over benchmarks to customers who follow the investment advice provided by the system total approximately US$244 million, while adding more than US$31 million to the bank's annual income.

"Opti-Money at Bank Hapoalim: A Model-Based Investment Decision-Support System for Individual Customers," Interfaces, 34(1): 39–50, 2004

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Nonlinear Optimization: Real-Life Applications

DHL:

- German logistics company
- employs more than 510,000 people
- operates in more than 220 countries
- moves 5% of the global trade volume



Major hurtle is to understand the customer behavior. They started using optimization based learning since 2004. The tool allocate marketing resources to activities that grow the global brand in their country.

The tool led to an increase in brand value of $1.32 billion over 5 years.

- corresponds to a return on investment of 38%
- the implementation let to major impact on DHL's strategy and organization

"Managing Global Brand Investments at DHL," Interfaces, 41(1): 35–50, 2010

Nonlinear Optimization: Real-Life Applications

Carlson Rezidor Hotel Group:

- the world's 9th largest hotel group by revenue
- 1,300 hotels in operation
- employs over 80,000 people

Revenue management is the art and science of selling the right product to the right customer at the right time for the right price.



Implement a revenue optimization project, Stay Night Automated Pricing (SNAP), started with enterprise demand forecasting across 600 US hotels in 2007.

- increased revenue by more than $16 million annually
- plans to deploy tool in Europe, Middle East, Africa, and Asia Pacific regions
- anticipates that the worldwide incremental revenue in excess of $30 million annually.

"Carlson Rezidor Hotel Group Maximizes Revenue Through Improved Demand Management and Price Optimization," Interfaces, 43(1): 21–36, 2013

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Nonlinear Optimization: Software Packages



Coin-OR
IPOPT



Convex Quadratic Optimization problems



IBM ILOG CPLEX





BARON



Xpress Solver

Convex Conic Optimization problems



open source → **Excel add-ins** → **high-end solvers**

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